


# *"SPILLOVER* EFFECTS OF FINANCIAL INCLUSION ON CONSUMPTION PATTERNS IN RURAL PERUVIAN FAMILIES"

**Research paper submitted**
**in fulfillment of the requirements for the degree of**
**Master in Economics**

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**2019**

Many thanks to our parents, Juan, Beatriz, Adrián and Isabel, for their support throughout this process. We also thank Bruno Saavedra for keeping us focused and caring about our progress during the busiest stages. We could not have done it without you; this work is by you and for you.

We are very grateful to Francisco Galarza for his valuable insights, ideas and feedback, which shaped the present version of this work. We also thank Nelson Ramírez for challenging our ideas and encouraging us to explore.

**Executive summary**

Financial inclusion allows households to access financial products to increase their accumulation of durable goods and long-term assets, so that a higher level of well-being is achieved and the reduction of poverty and inequality is ensured. However, in Peru most districts do not have a financial system service point within their territory. Given this situation, it is relevant to analyze whether the physical distance from households without direct financial access to another district that does have a financial presence generates some kind of *spillover* effect on the consumption patterns of families in rural areas of Peru.

Accordingly, this document proposes pseudo-panel models with instrumental variables to evaluate this effect over the 2010-2017 period, using financial access and usage variables from the Superintendency of Banking, Insurance and Private Pension Funds (SBS) and data on the composition of household expenditure from the National Household Survey on Living Conditions and Poverty (Enaho). To identify this *spillover* effect, dummy variables were generated that classified families unserved by the financial system according to the distance between a district without financial supply and the nearest district with at least one banking agent (cajero corresponsal).

The results show that, despite not having direct access to the financial system, physical proximity to districts with financial supply has a positive effect on the share of expenditure allocated to durable goods, relative to families located more than 50 km from a territory with a financial presence. Moreover, this effect grows as the distance shrinks. On the other hand, those located more than 50 km away are found to incur higher total expenditure than unserved families located closer to the financial system, although when the distance is short the difference relative to the most distant families is smaller. Therefore, based on the results, even though rural towns have no direct financial access or usage, their proximity to other territories that do have a financial presence modifies consumption patterns in a robust and significant way relative to the rest of the families not served by the financial system.

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## Chapter I. Introduction

At the macroeconomic level, the literature shows a positive effect of financial development on increasing countries' growth rates and on reducing both inequality and poverty. Therefore, governments and international organizations have set the goal of ensuring financial inclusion among low-income sectors so that they too can enjoy the benefits associated with development (United Nations [UN] 2015). In this context, there is great interest in identifying which individual characteristics predict access to and use of financial services.

In Peru, because growth in financial demand has been lower than that of financial supply over the last ten years, several recent studies have focused on identifying the key household characteristics or components that affect participation in the financial system (Jaramillo *et al.* 2013, Cámara and Tuesta 2015, SBS 2018). Although these studies seek to mitigate low levels of financial access on the demand side, they do not deepen the discussion on the next stage of the problem: is there a positive and significant effect of this greater access on household well-being?

Theoretically, the consumption of financial products allows families to increase their holdings of physical and human assets without compromising their current expenditure needs (Banerjee *et al.* 2015). However, the recent literature on the Peruvian context has not been able to show that greater financial access, or financial inclusion in general, has an effect on people's quality of life in terms of well-being. It is particularly important to carry out an analysis of this kind on low-income populations in rural areas, since they are the segment of the population most underserved by the financial system, and access would allow them to increase the accumulation of their productive, educational and health assets from their low initial levels.

In the context of low-income families, there is a wide range of studies evaluating the effectiveness of various targeted credit programs in achieving a positive impact on certain well-being measures for this segment of the population (Pitt and Khandker 1998, Monge-González *et al.* 2007, Karlan and Zinman 2009). However, due to the different conditions or settings in which these programs are implemented, the current evidence on participation in such programs is inconclusive as to the magnitude and direction of the resulting effect. Furthermore, since most of these studies are randomized controlled trials carried out in contexts different from Peru's, their findings are hardly applicable to the Peruvian reality, even in rural areas.

Despite the exponential expansion of financial supply since the early 2010s, at the end of 2017, 64.9% of districts still had no physical presence of the financial system in any of its forms (Superintendency of Banking, Insurance and Private Pension Funds [SBS] 2019). It is worth mentioning that all unserved localities could be considered rural areas given their location and number of inhabitants. Nevertheless, some districts without direct financial access could be exposed to a *spillover* effect, resulting from their physical proximity to other districts with a financial presence in their territory, that positively impacts some measure of well-being of their inhabitants, who are low-income rural families.

Given the literature's current interest in identifying the channels through which the development and well-being of low-income households can be ensured, it is important to evaluate the impact of the deployment of Peruvian financial system service points by financial institutions, and of the greater use of financial services, in the rural context. In particular, this research focuses on quantifying whether the *spillover* effect generates a positive and significant impact on two expenditure-related well-being measures: the share of expenditure allocated to the purchase of durable goods or long-term assets within the consumption basket, and total household expenditure. Taken together, both measures provide a complete picture of how financial inclusion affects consumption patterns, by identifying changes in the reallocation of expenditure and in total expenditure.

In order to analyze the effect of financial inclusion on the well-being of rural families, this study uses information on the composition of household expenditure from Peru's National Household Survey (Enaho) and statistics on the level of financial access and usage from the SBS. With these data, pseudo-panel models with instrumental variables are proposed to evaluate how the expansion of financial system service points has affected consumption patterns in population centers of no more than 4,000 households without a physical financial presence between 2010 and 2017. To estimate this, we used the distance in kilometers from the district where households without a physical financial system presence are located to the nearest district with at least one banking agent (the type of financial supply with the greatest penetration in Peru).

The results show that a physical distance of less than 50 km from the nearest district with a banking agent has a positive and significant effect on the share of expenditure allocated to durable goods, relative to families living in an unserved district located more than 50 km away. Moreover, this effect grows as the distance shrinks. Thus, the coefficient for districts less than 20 km away is almost double that of living between 20 km and 50 km from the nearest district with a financial

presence.

In contrast, the opposite effect is found for total household expenditure. The results show negative and significant coefficients for a physical distance of less than 50 km to the nearest district with a banking agent on the total expenditure measure, relative to families living in an unserved district located more than 50 km away. However, the magnitude of the negative effect on total expenditure of being less than 20 km away is smaller in absolute value than that of living between 20 and 50 km away.

Therefore, families located more than 50 km away spend more than those less than 50 km away, but probably on non-durable goods. This finding suggests that greater distance from the financial system may be related to a subsistence-oriented allocation of resources. Accordingly, we propose improving connectivity between rural districts so that the costs of accessing the financial system decrease and the reach of *spillover* effects can be increased.

The study is structured as follows. Chapter II introduces the concepts associated with financial inclusion in order to review the main international findings on the impact of microcredit on various well-being measures. The same chapter presents the situation of the Peruvian financial system over 2009-2017 and reviews the main studies on financial inclusion in Peru. Chapter III then sets out a theoretical framework for understanding the effect of loans on household well-being, reviews the information sources, and presents the models used to analyze the effects of financial access and usage. After presenting the hypotheses, Chapter IV shows the main regression results. Chapter V then contrasts the results with findings from the literature and sets out next steps for future similar research, along with policy recommendations. Finally, the conclusions are presented.

## Chapter II. Literature review

Much of the recent international literature on financial inclusion has focused on finding the determinants of financial access, since access allows people to overcome their budget constraints, smooth intertemporal consumption and mitigate the negative effect of unanticipated financial shocks (Demirgüç-Kunt *et al.* 2008, Beck *et al.* 2008). However, the concept of financial inclusion encompasses much more than just access to the financial system.

According to Demirgüç-Kunt and Klapper (2012), financial inclusion enables affordable, timely and adequate access to financial services, broadening their use with the purpose of promoting financial well-being and economic and social inclusion across all segments of society. Based on this concept, three key dimensions of financial inclusion can be distinguished: access, usage and well-being. These three concepts are defined below based on the existing literature.

### 1. What do we mean by financial inclusion?

Access to financial services is understood as the availability of quality financial service points at an affordable price that allows people to use financial services. Therefore, households' physical proximity to financial system branches and the absence of monetary and non-monetary obstacles to using the services determine financial accessibility (World Bank 2006, Claessens 2006).

Additionally, more recent definitions of financial access encompass the convenience and willingness of potential users to employ the products and services offered. Therefore, this concept relates to the ease of using financial system products, regardless of whether people decide to use them or not (Ellis *et al.* 2010). Thus, an underlying demand for financial services is not assumed for all individuals in proximity to financial supply (Kostov *et al.* 2012).

Usage, on the other hand, refers to the effective and frequent consumption of financial services over long periods. This implies that, under a standard supply-and-demand framework, financial access determines the supply and demand curves, while usage would be found at the intersection of these curves (Claessens 2006). However, the use of financial services is subject to several supply factors related to information asymmetries (moral hazard and adverse selection), contractual problems, transaction costs, institutional factors or even country-level characteristics (Beck and De la Torre 2007). These market failures cause localities with similar financial supply to show different levels of effective usage.

Regarding the estimation of financial service usage, although each individual in the same locality consumes financial products and services at a different level, measurement is usually carried out at the aggregate level. Additionally, usage is usually distinguished by the products offered in the market, since each one meets different customer needs, especially in the context of poor and remote localities. For example, a high volume of loans indicates a greater need for liquidity, while the analogue in deposits could indicate preferences for intertemporal consumption smoothing.

Therefore, financial access and usage are closely related. In short, the availability of financial service points alone is not enough to achieve the inclusion of communities. However, this measure of access does turn out to be a necessary condition for people to use financial services, although the level of usage depends more on the financial needs the population seeks to meet (Talledo 2015).

Finally, the benefit is the effect on individuals' quality of life resulting from greater financial inclusion. To observe this benefit, it is necessary to quantify the change in some measure of well-being as a consequence of an increase in financial access and usage variables (Sarmiento 2018). Thus, because financial system products make it possible to increase holdings of physical and human assets without affecting regular expenditure levels, the success of financial inclusion depends on household development through the simultaneous accumulation of long-term assets and the satisfaction of their expenditure needs (Townsend and Ueda 2006, Agnello *et al.* 2012, Cull *et al.* 2014).

This objective is even more important in the context of low-income populations, since financial inclusion allows them to reduce the high variability of their consumption by giving them mechanisms to efficiently cope with adverse income and price shocks. It also allows them to increase the value of their productive, educational and health assets, with macroeconomic implications through positive impacts on economic growth and on the reduction of inequality and poverty (Honohan 2008, Sarmiento 2018).

Unfortunately, as mentioned above, the literature has focused more on studying the demand-side determinants of access to the financial system, neglecting the analysis of the effect of greater access and usage on household well-being. Furthermore, although there are country-level studies of financial inclusion, different effects may be observed at a more disaggregated level, such as the household. For example, Beck *et al.* (2012) find that credit is positively related to economic

growth and to the accelerated reduction of inequality, but when distinguishing between loans to firms and to households, this effect is seen to result solely from business activity, since loans directed to households show none of these effects.

On the other hand, given the positive effect of financial inclusion on low-income populations, it is important to analyze whether the expansion of physical financial supply affects the well-being of families that lack direct access to the financial system, depending on how underserved they are. Although the effect of physical financial presence has been analyzed, there is no in-depth analysis of this potential indirect effect. Therefore, this work aims to verify whether the current provision of financial services in Peru has managed to increase any measure of well-being for rural Peruvian families not served by the financial system.

To analyze rural families without direct access to the financial system, their well-being must be compared with that of families in a similar context that do access and use the financial system. Therefore, we review the microcredit literature since, although microcredit aims to strengthen productive capacity, it is a tool that has been studied in depth in order to see whether it has an impact on the well-being of the users or families that receive it.

## 2. International evidence on the impact of microcredit on benefits in vulnerable sectors

Recently, microfinance has attracted a great deal of attention from researchers because of its widespread perception as an effective tool for improving social conditions in developing countries (Bebczuk and Haimovich 2007). Hence, the rapid spread of microcredit use over the last decade is not surprising, since many people and institutions believe it is a successful initiative in the fight against poverty (Banerjee 2013).

According to Schroeder (2012), the belief that microcredit is a powerful tool for alleviating poverty is founded on its nature of providing small amounts of liquidity to the poorest so they can undertake business ventures. Therefore, in theory, microcredit allows households to acquire durable goods and engage in income-generating activities that they could not undertake without financial support, being limited by their purchasing power. If this theoretical premise holds, the empirical evidence should reflect a sustained increase in long-term[1] benefits. However, to date, the empirical evidence on the poverty-mitigating and benefit-generating effect of microfinance is ambiguous and insufficient, even though a significant effect is assumed to exist (Schroeder 2012).

[1] Much of the microfinance literature uses a consumption variable rather than income levels as a measure of well-being. This is because financial inclusion is studied in developing countries, where the poverty line is usually quantified in terms of consumption baskets (Schroeder 2012). In addition, income data reported in longitudinal surveys are usually subject to various errors (Deaton 2018).

One of the most significant studies on the importance of microcredit is a quasi-experimental study by Pitt and Khandker (1998) that analyzes the effect of credit programs and financial services on resource allocation in rural households in Bangladesh between 1991 and 1992. The study finds that access to credit programs positively affects schooling and increases labor supply, asset holdings and household expenditure when the recipient of the service is a woman. This finding raises the idea that loans granted to men are related to higher spending on leisure.

To complement these findings, Schroeder (2012) analyzes the same database of rural villages in Bangladesh but adds a time component to the model, using the follow-up survey carried out between 1998 and 1999. With this information, he seeks to show that the effects of microcredit loans may not be immediately evident, but rather that the benefits are really observed in the long run. Thus, after controlling for the endogeneity of the decision to take a loan, Schroeder finds that microcredit has a positive and significant effect on long-run consumption. Furthermore, the elasticity of household consumption with respect to loans is estimated to lie between 0.193 and 0.212 for a random household in the sample.

Additionally, Khandker (2005) analyzes the impact of credit programs and their *spillovers* on consumption and poverty reduction using the same database as Schroeder (2012). The study finds that loans to women increase household consumption, while loans in general generate a substantial reduction in poverty. The effects on poverty are larger for both moderate and extreme poverty. Both results are observed among participants and non-participants in the credit programs, even in areas where these programs are not present. This finding suggests the existence of *spillover* effects due to proximity to program participants, at least for families affected by moderate poverty. Finally, the results also indicate that more than half of the 3.0 percentage points of annual poverty reduction observed for participants is a direct consequence of microcredit.

In addition to reducing poverty, microcredit can play a risk-mitigating role. According to Gertler *et al.* (2009), given the public sector's limitations in providing appropriate health care nationwide and the high private costs of medical services, microcredit may make it possible to insure the health of the most vulnerable populations without affecting their current consumption.

In line with this idea, Islam and Maitra (2012) analyze the effect of access to microcredit on consumption variability in the face of health *shocks* using a panel of individuals from rural Bangladesh between 1997 and 2005. The study shows that, in the long run, households with

access to credit that suffered a negative health *shock* did not reduce their livestock holdings, whereas households in the control group did. Thus, these findings reinforce that, with greater access to credit, household members are better able to protect their asset holdings and their levels of food and non-food expenditure, reducing the variability in the acquisition of durable and non-durable goods.

Although much of the literature has focused on Asian countries, positive effects of microcredit have also been found in Latin America. This region presents a different macroeconomic and social context, with a tendency toward high growth rates with large annual variability. Hence, Bittencourt (2012) proposes that the Latin American environment offers a suitable setting for testing the validity of Schumpeter's model, which states that credit in the hands of a skilled entrepreneur leads to innovation and prosperity.

To test that financial development fosters economic growth through innovation and entrepreneurship, Bittencourt (2012) uses macroeconomic panel data for Argentina, Bolivia, Brazil and Peru between 1980 and 2007. After controlling for inflation and the other political and economic conditions that affected the countries' performance during the period, the study validates Schumpeter's premise that credit generates innovation and prosperity, and argues that the effect of credit is greater when governments ensure conditions conducive to the development of the financial industry.

Truly validating the premise of Latin American entrepreneurship requires deepening the analysis to the microeconomic level to evaluate the implications of financial development for household well-being. Along these lines, Monge-González *et al.* (2007) analyzed the effect of microcredit granted by the Banco Nacional de Costa Rica to small and micro enterprises on the productivity of their operations and on the well-being of the low-income employees working at those firms. Ordinary Least Squares (OLS) estimates show that credit has a significant and positive effect on productivity, both directly and indirectly through innovation and capital accumulation. Additionally, families working for small and micro enterprises that receive loans from the Banco Nacional are found to have higher household income than households of employees at firms that are not clients of that bank.

In another microeconomic study, Bebczuk and Haimovich (2007) analyze the impact of credit on the income of poor families and their schooling decisions in several Latin American and

Caribbean countries[2]. The estimates of both effects were positive and varied in magnitude across countries, but were not significant in all cases. Moreover, due to data limitations, in most cases only the effect of receiving a loan as reported by the household head was evaluated, which does not allow controlling for other important variables, such as the quality of the credit, the financial institution offering it, or the repayment conditions offered to the head in terms of interest rates and collateral.

Particularly in Latin America and the Caribbean, one criticism of microfinance is that loan interest rates tend to be far above what low-income people can afford. Although average interest rates in the region fell by 0.9% annually between 2003 and 2006, effective interest rates remain above 25% (Rosenberg *et al.* 2009). However, financial institutions justify these rates by the high costs associated with monitoring borrowers and the need to hedge against the risk of non-repayment.

Against this backdrop, Bogan *et al.* (2015) set out to analyze the characteristics that affect the price elasticity of demand for microcredit. To do so, they used responses from clients of a microfinance institution in the Dominican Republic on the amount they would be willing to borrow in response to changes in interest rates. The findings show that an entrepreneurial attitude and lower risk aversion are related to elastic demand, while greater financial literacy is associated with inelastic demand. These results show that it is possible to appropriately segment the interest rates offered to different types of clients in order to ensure greater financial access among low-income populations and thus achieve greater coverage and well-being effects.

As can be seen, there is a vast literature emphasizing the existence of a positive impact of microcredit on households, whether direct or indirect. However, there are also studies that identify contradictions and a lack of statistical evidence to validate the effectiveness of microcredit on household well-being (Roodman and Morduch 2014). For example, Morduch (1998) criticizes the use of regression discontinuity because the theoretical eligibility cutoff for receiving a loan in Bangladesh was not met in practice, so the division between treatment and control groups loses validity. Replicating the study by Pitt and Khandker (1998) with a different methodology, Morduch (1998) reports no significant differences in increased consumption and schooling relative to non-participants, but does confirm that access to credit reduces consumption variability through income smoothing.

[2] The data consolidate information from household surveys from Bolivia, Guatemala, Jamaica, Mexico, Nicaragua, Peru, Paraguay and Haiti (Bebczuk and Haimovich 2007).

On the other hand, Coleman (1999) analyzes the effect of group lending by "village banks"[3] on 72 development variables for villages in northern Thailand between 1995 and 1996, but finds non-significant impacts on physical asset holdings, savings, production, sales, productive expenditure, time devoted to work, and part of the expenditure on education and health. Given these negative results, he proposed several hypotheses about their causes; for example, that the loan amount was insufficient to promote productive activities, that interest rates were high relative to the communities' circumstances, or that the focus of these loans may have been consumption stability.

To test these hypotheses and explore potential differentiated effects, Coleman (2006) studies the same villages in Thailand to analyze whether there are significant differences in the impact of credit depending on participants' wealth. The results show that the wealthiest individuals in the village are more likely to participate in the "village bank" group lending program and that this wealthier group takes the largest share of the loans. Given this, it is not surprising that credit for the wealthy group generates large effects on all the well-being measures that had been non-significant in his previous research. He thus concludes that microcredit can help some poor families but fails to meet the needs of the most vulnerable groups, reinforcing the finding of Amin *et al.* (2003).

In another study on targeted credit, Banerjee *et al.* (2015) conducted a randomized experiment on microcredit in low-income localities in India where there had previously been no microfinance presence. Eighteen months after the intervention, no significant differences in expenditure levels were observed relative to the control groups. However, there was a change in the composition of expenditure, as households increased their consumption of durable goods and reduced their spending on leisure ("temptation") goods. Regarding effects on labor productivity, the authors highlight greater allocation of resources to own businesses in the treatment groups with access to credit. Nevertheless, none of these positive effects had a significant average impact; even the impacts of microcredit on health, education and women's empowerment were not significant.

Given this result, the study by Karlan and Zinman (2009) on the use of consumer credit is interesting because, by focusing the financial service on spending on goods and services, this type of loan is unproductive from a theoretical point of view. The authors studied people whose credit applications had been rejected by a South African bank that grants high-yield, high-risk loans without collateral. From this sample of rejected applicants, a consumer credit line at that bank was offered to a random group. The findings show that access to this type of credit had a positive and

[3] "Village banks" are made up of groups of 20 to 60 people, usually women, who receive money from an NGO for 6 months and guarantee one another to ensure repayment to the institution (Coleman 1999).

significant impact on job retention, income, food consumption in terms of both quality and quantity, and the prevention of adverse events. Finally, the bank made a profit on the loans granted, suggesting that consumer credit has the potential to generate social and economic well-being in the population.

As can be seen, most microcredit studies analyze effects through controlled experiments. However, it is also important to evaluate the effect of the natural evolution of financial supply on the well-being measures of different families, with and without direct access to financial service points, to see whether there is indeed a change at the national level.

In the case of the most vulnerable families living in remote areas without financial service points (the population under analysis), it is relevant to analyze whether physical proximity to the financial system has any effect on the well-being of this segment of the population. Therefore, we present the existing literature on the effect of financial expansion, physical proximity to the financial system, and indirect financial access.

In the context of Bangladesh, Ahmed (2016) uses household data by district to estimate consumption premiums that show whether district expenditure is above or below average. He thus evaluates the impact of the direct physical presence of microcredit institutions on these relative consumption measures between 2005 and 2010. The study finds that a household without loans is positively affected by the presence of a microfinance institution. Moreover, owing to the expansion of financial supply in low-income sectors, households that do not use financial services still benefit from the presence of banks nearby, due to the growth in economic activity.

In India, on the other hand, Burgess and Pande (2005) analyzed the effects of a state bank branch expansion program between 1969 and 1990. During this period, around 30,000 financial service points were opened in rural areas that previously had no direct access to financial services. The results indicate that, as a result of the expansion of the rural network, the presence of financial service points significantly reduced the share of people below the poverty line at the state level through greater use of deposits and credit in rural areas.

However, it is important to note that this study uses states as the unit of analysis, since its well-being measure is the state poverty rate (Burgess and Pande 2005). It is therefore not possible to discern the effect on families living in different districts within a state, nor to distinguish whether

the reductions in state poverty result from increased access to and use of financial services or from *spillover* effects on families living in territories without a direct financial presence.

In the case of Mexico, Bruhn and Love (2009) analyze the effect of the expansion of Banco Azteca branches, which focus on providing financial products and services to low-income people, on aggregate well-being and economic activity measures. The increase in financial access raised the number of informal businesses and income levels and reduced unemployment compared with municipalities without a Banco Azteca presence. However, the authors acknowledge that their findings do not distinguish whether these effects were direct, that is, the result of greater use of financial services, or whether the expansion generated greater competition, which indirectly affected Mexico's economic activity.

Taken together, the above results show the current debate on the effectiveness of credit for household well-being across different sectors and under different types of credit and contexts. Given the limited literature on the effect of the expansion of physical financial supply on household-level well-being measures, it is important to deepen the analysis and generate greater dialogue about the implications for underserved low-income populations. In the Peruvian case, there are no studies covering this effect of bank network expansion, but there are some microeconomic studies that analyze the determinants and effectiveness of financial inclusion. To contextualize these studies, it is important to first describe the characteristics of the Peruvian financial system during the period of analysis.

## 3. The financial system in Peru

In Peru, the formal financial system is made up of the institutions supervised by the SBS. This regulator constantly supervises the operations of the different types of financial institutions nationwide, which, as of April 2019, can be divided into 17 banks, 11 finance companies, 11 municipal savings and credit banks (CMAC), 6 rural savings and credit banks (CRAC), 9 small and micro enterprise development entities (Edpyme), and other state entities or specialized companies (Central Reserve Bank of Peru [BCRP] 2019). Additionally, all these institutions can be divided into two groups: multiple banking institutions (made up of banks and some finance companies) and microfinance institutions (MFIs) (made up of CMACs, CRACs, Edpymes and some finance companies specialized in microfinance). These financial institutions differ in terms of number of branches, target clients, relationship with the central bank, ownership structure and other aspects (Aguilar 2016).

Multiple banking institutions are those that use money received from the public and other funding sources to carry out investments and banking operations subject to market risk, and to provide credit and loans (Lizarzaburu 2014). As of 2017, banks held 89.35% of financial assets nationwide. Furthermore, although the share of branches in Lima decreases every year, 54.89% of all bank branches are still located in Lima, and they concentrate 85.42% of deposits and provide 78.56% of national bank loans (SBS 2019). Therefore, a large part of the services offered by multiple banking is centralized in the capital.

MFIs, on the other hand, have managed to expand the supply of financial services to the lowest-income segments of the population, with special emphasis on providing personalized loans to finance personal projects and activities of low-income individuals (Aguilar 2011). The most important representatives of these institutions are the CMACs, which belong to municipal governments and operate under a decentralized model to strengthen opportunities for financial usage among vulnerable populations, using cash-flow analysis instead of the collateral-based methodology employed by multiple banking to assess their clients' creditworthiness (Jaramillo 2014).

It is worth noting that CMACs initially operated locally, but since 2002 they have been authorized to operate throughout the national territory. Meanwhile, banks have acquired various specialized financial institutions to meet the needs of this market niche, and CMACs have managed to expand their range of financial services to serve more people (Aguilar 2016). As a result, greater competition has emerged between both types of financial institutions, especially in microfinance.

In addition to acquiring microfinance institutions, multiple banking institutions have bet on expanding their coverage by massively rolling out service points in remote areas. To this end, they have promoted a service channel called the banking agent (cajero corresponsal), which makes it possible to make deposits and payments and purchase insurance outside bank branches. This channel of the financial system is mostly located within commercial establishments, such as small grocery stores and pharmacies in urban and rural areas, complementing the supply of ATMs located inside banks or in large shopping centers (Cámara and Tuesta 2015). It can thus be stated that banking agents and CMACs together seek to foster financial inclusion among poor households and micro and small enterprises (Mypes), so that they can generate significant improvements in well-being for society.

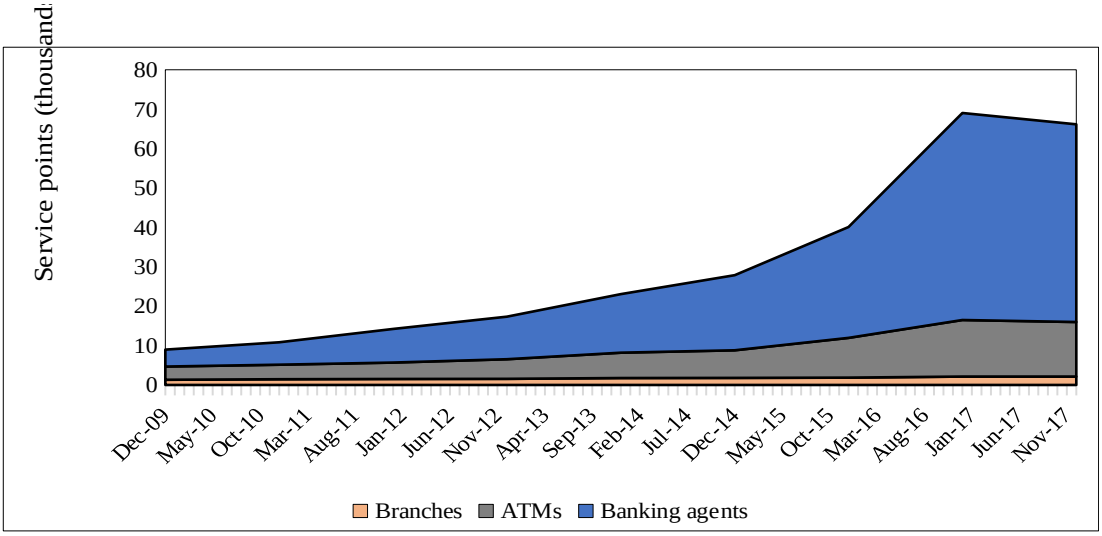
The remaining MFIs play a smaller role in increasing competition nationwide. CRACs began as an option targeting the rural and agricultural sector for the provision of financial resources, and to date they have a smaller share than CMACs. Edpymes, for their part, are formal NGOs licensed to run microcredit programs focused on low-income clients, such as small producers and traders in semi-urban and rural areas (Aguilar 2016). Since Edpymes are not authorized to take deposits from the public and must be funded through credit lines, several have been absorbed by commercial banks because of their knowledge of microfinance and their high growth in the country (Cotler and Aguilar 2013).

Owing to their characteristics, MFIs have achieved greater penetration in territories outside the capital. For example, according to SBS statistical reports for 2017, 81.06% of all CMAC branches are located outside Lima, and they concentrate 69.63% of all national CMAC deposits and provide 84.76% of all national CMAC loans. However, despite the high decentralization of their services, they account for only 6.20% of national assets (SBS 2019). Thus, although MFIs have achieved a larger presence in the poorest sectors, their aggregate concentration of funds nationwide remains far below that of multiple banking.

In recent years, private and public financial system companies have made a great effort to expand their networks of service points in absolute terms. Thus, between 2009 and 2017, the period of analysis of this research, basic bank financial supply, which considers the total number of service points[4] in multiple banking offering savings, deposit and credit services, increased sixfold nationwide. This growth is due to the exponential development of banking agents, although their upward trend was interrupted in 2017 (see figure 1).

[4] In Peru, the service points within basic bank financial supply are the number of bank branches, the number of ATMs and the number of banking agents.

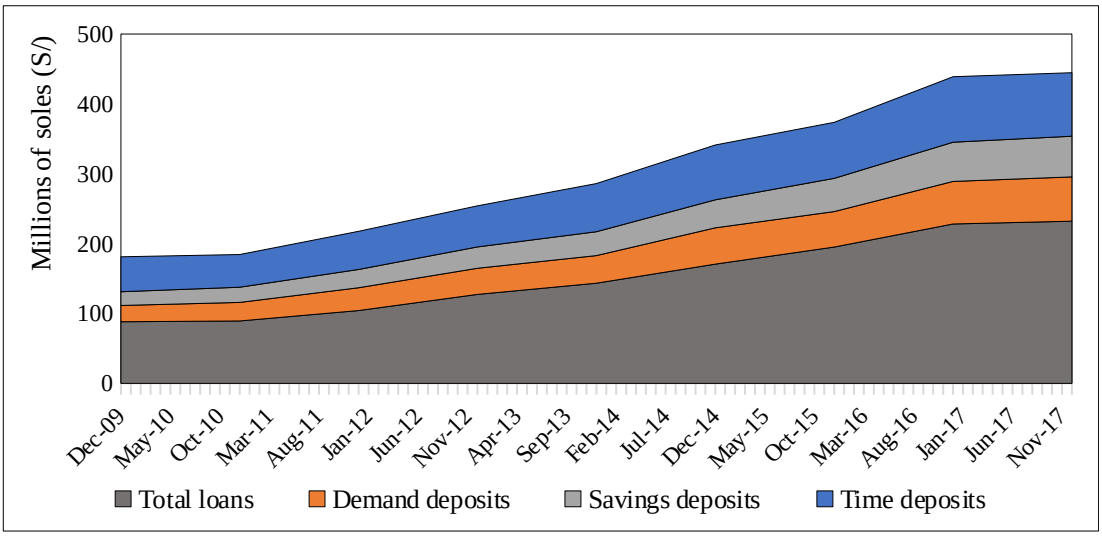
**Figure 1. Growth of banking financial service points nationwide (2009 - 2017)**




Source: SBS 2019. Authors' elaboration

Additionally, the value of financial services provided by multiple banking institutions increased by 145.57% between 2009 and 2017. Figure 2 shows how the use of bank financial services is distributed between credit and the different types of deposits according to their liquidity.

**Figure 2. Growth of multiple banking assets nationwide (2009 - 2017)**



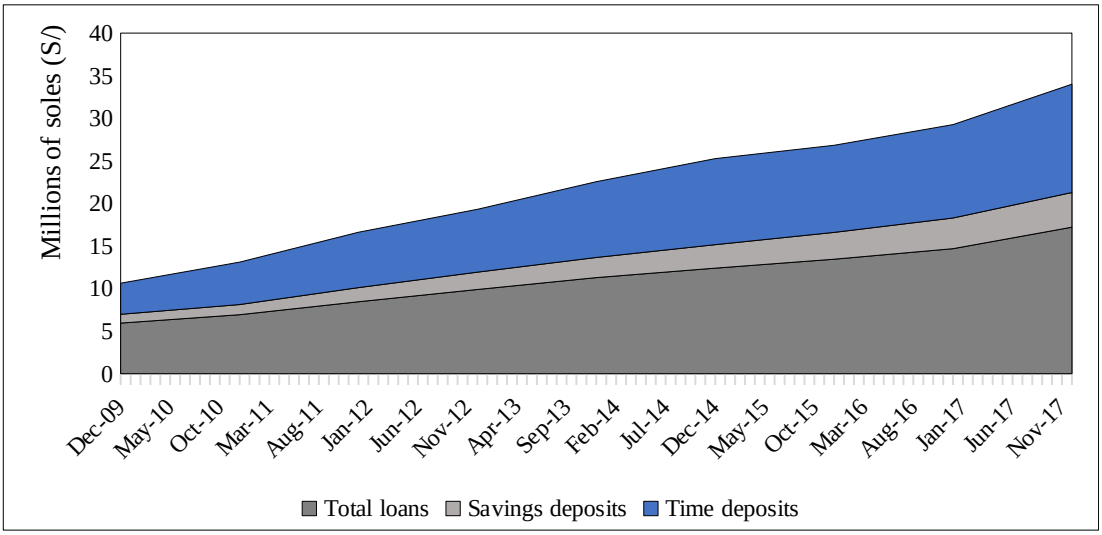

Source: SBS 2019. Authors' elaboration

In the case of CMACs, between 2009 and 2017 service points grew from 346 to 750 branches nationwide; this type of institution therefore saw more modest growth of 116.76%. However, the

value of CMAC financial services increased proportionally more than that of multiple banking institutions, growing by 220.26% between 2009 and 2017.

Figure 3 shows how the use of services is distributed between credit and the different types of deposits according to their liquidity[5]. Thus, although banks have achieved greater physical expansion across Peru, CMACs have grown faster in terms of financial usage.

**Figure 3. Growth of municipal savings bank (caja municipal) assets nationwide (2009 - 2017)**

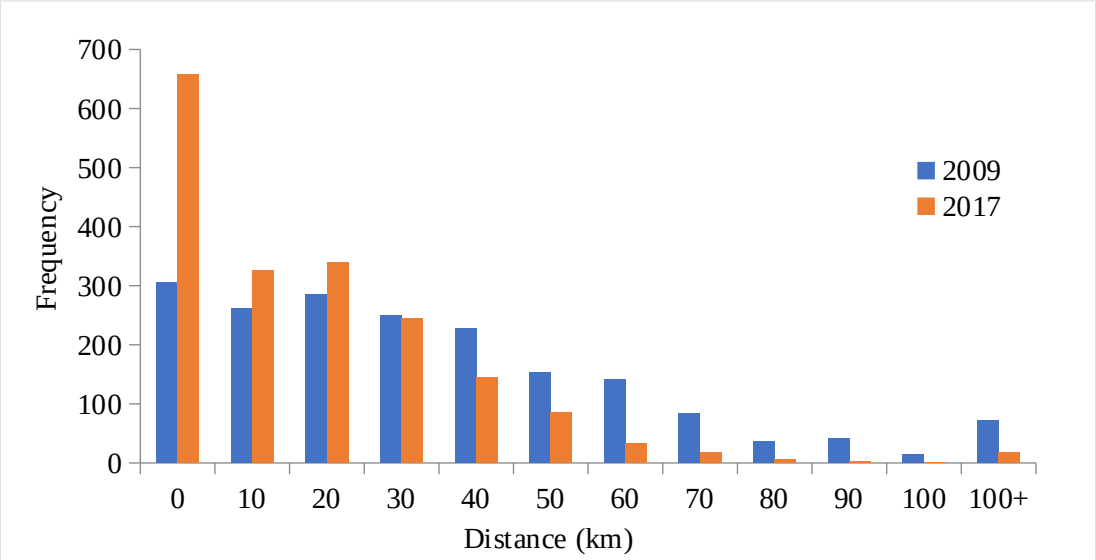



Source: SBS 2019. Authors' elaboration

However, this exponential physical expansion of financial supply has not achieved nationwide coverage by the financial system. Figure 4 shows how the number of districts not served by at least one banking agent evolved between 2009 and 2017.

[5] Unlike multiple banking, municipal savings banks do not offer demand deposits as a financial product.

**Figure 4. Evolution of the distances from Peru's districts to the nearest district with at least one banking agent, 2009-2017**




Source: SBS 2019. Authors' elaboration

At the beginning of the period of analysis, 83.71% of districts had no physical financial presence and 16.5% of districts were more than 50 km away from a district with a banking agent. By 2017, unserved districts had fallen to 62.9%, still a significant percentage. Additionally, only 4% of districts were more than 50 km away from a district with a financial presence, and distances in general can be seen to have decreased. Therefore, there is still room to close financial connectivity gaps.

Regarding financial usage, on the other hand, multiple banking and CMACs concentrate more than 70% of their loan portfolios in the least poor quintile of the population (León 2018). Furthermore, the number of borrowers nationwide increased by only 53% between 2009 and 2017. This could mean that the wealthiest segment of the population has increased the frequency and amount of its loans and deposits, so there is still no widespread growth in financial usage or access across the bulk of the Peruvian population. Thus, although Peru shows some progress in access to financial services on the supply side, the promotion of access and usage on the demand side still appears limited at the aggregate level.

However, analyzing disaggregated data on clients nationwide reveals a substantial increase in the number of small loans to both individuals and legal entities. According to the SBS (2018), the

number of borrowers owing less than S/ 9,000 who are legal entities increased by 50% between 2013 and 2018, while the number of borrowers owing less than S/ 3,000 who are individuals rose by 29.95% over the same period. Therefore, the lower-income population is more easily accessing funds that could mean a future increase in well-being, which varies depending on the type of client of the financial institution[6].

Despite this growth, it must be acknowledged that, in Peru's low-income segments, the interest charged by financial institutions on consumer and Mype loans is high, due to the high cost and risk[7] firms face when disbursing their funds (Choy *et al.* 2015). Therefore, part of the low-income population cannot use formal financial supply because it is too expensive and the requirements demanded by financial institutions are not suited to their conditions (Alvarado *et al.* 2001).

Faced with this situation, and given that families' ultimate goal is economic progress, low-income households turn to informal credit, a more accessible short-term means that responds only to market conditions within the small geographic area in which it operates (Alvarado *et al.* 2001). The most common forms of informal financing include loans from relatives, friends, traders, pawnshops, businesses, landowners and rotating savings groups (juntas). Although these lenders have a greater presence in the poorest strata, they still require collateral to grant loans. The difference from formal financial supply is that informal lenders, not being supervised by a regulator, accept a much wider range of collateral[8] (Trivelli 2000 and 2002).

Regardless of whether financial supply is formal or informal, all credit demand is a derived demand, since credit is not demanded for its own sake, but to finance production, consumption or investment needs. In low-income strata, there is no clear separation between consumption and production activities, so informal credit serves to finance short-term expenditure on durable goods, non-durable goods, education, health and other productive expenses (Alvarado *et al.* 2001).

The lack of reliable national data on informal loans is a limitation for studying the formal financial system in areas where informality exists, since formal access and usage measures will necessarily capture the effect of the informal market. The expansion of formal financial supply described

[6] For households, financial inclusion is a tool to stabilize consumption and cope with economic shocks, while for firms it means financing opportunities and productivity improvements in operations (León 2018).
[7] Institutions invest in monitoring their clients to ensure repayment and, in the event of default, must bear the loss (Choy *et al.* 2015).
[8] The most popular forms of collateral among informal lenders are appliances, vehicles and intangible collateral; the latter corresponds to loans made to acquaintances, where the collateral is the trust placed in the borrower (Trivelli 2000).

above would therefore be overestimated. Problems associated with the degree of formality of the financial system, which affect the estimation of the model, are discussed and addressed in greater depth in the section on information sources and data.

Given this landscape of the Peruvian financial system, the formal and informal financial systems together may be fulfilling their role of efficiently channeling resources to low-income households. Unfortunately, according to Honohan (2008), greater use of financial services may have only a slight effect on well-being and on reducing poverty and inequality, since people with little financial literacy are more prone to making poor investment and intertemporal consumption decisions. It could therefore also be that no productive expenditure is generated that would allow *spillover* effects on families living in areas without a physical financial presence. However, to confirm these premises, it is necessary to observe the effect on the quality of life or well-being of these families.

As mentioned above, there are no studies in Peru investigating the effect of bank network expansion on household well-being, but there is research analyzing the determinants and effectiveness of financial usage. The main findings of these studies conducted in the Peruvian context are therefore presented below.

## 4. Financial inclusion in Peru

Although the Peruvian economy has experienced sustained growth for several years, SBS evidence (2018) on the evolution of national financial inclusion variables shows that the expansion of the financial system has not achieved the expected level of penetration in the use of financial services. This problem is not new, and there have been several past efforts to identify the determinants of financial access in the Peruvian context in order to identify policies that could encourage the participation of the most vulnerable households in the financial system.

In the study by Cámara and Tuesta (2015), the authors analyzed the effect of a set of socioeconomic variables on the probability of belonging to the financial system. Using Enaho data, they constructed a dummy variable that took the value of 1 if the household had a mortgage, received interest from a financial product or carried out banking transactions, and 0 otherwise. This variable was therefore used as a *proxy* for citizens' bankarization. Their results showed that low levels of income and education, living in a rural area, and being a woman reduce the

probability of being included in the financial system. They also found evidence that the magnitudes of the effects differ depending on the region where families live.

Along the same lines, Jaramillo *et al.* (2013) studied the determinants of credit use with Enaho data. For this analysis, the total amount of credit at the district level from the SBS was used as the dependent variable, and its lag was used to incorporate credit history as a control. The results show findings similar to those of Cámara and Tuesta (2015) on the effects of the household head's education and sex, but highlight that, for the poorest sectors, access to adequate public service infrastructure[9] has a positive effect on holding credit. They also indicate that this segment of the population is negatively affected by the geographic conditions in which they live, since distance and terrain increase the cost of accessing the scarce formal financial system supply in the interior of the country.

Complementing these findings, Sotomayor *et al.* (2018) analyzed the effect of the supply and demand environmental factors[10] of the financial system on access to and use of financial services in Peru. To do so, the authors used data from the National Survey on Demand for Financial Services and Level of Financial Culture (ENDSF), which collects the financial habits, characteristics and needs of the Peruvian population. On the supply side, they find that shorter travel time to service points predicts greater use of deposit accounts, but that this time has no significant effect on holding a deposit account or on access to credit.

On the demand side, a higher income level turns out to be a relevant factor for holding and using deposit accounts, although this effect is not observed in rural areas. Although this study tries to distinguish the effects of financial culture and the use of public services on access and usage across different geographic strata, most of the results appear inconsistent, showing, for example, a positive effect on access but a negative one on usage. Therefore, according to Sotomayor *et al.* (2018), it is necessary to deepen the analysis of the dimensions of financial inclusion to understand the possible nuances between effects, whether opposite or in the same direction.

In the case of the rural sector, Alvarado and Pintado (2017) analyzed bankarization and credit supply in the agricultural sector, where a low level of access to financial services is observed despite a good disposition to access credit. Among the determinants found, the authors show that having a mother tongue other than Spanish has a negative effect on applying for and obtaining

[9] The infrastructure analyzed includes access to drinking water, sewerage, fixed and mobile telephony, and electricity (Cámara and Tuesta 2015).

[10] In the context of that research, environmental factors encompass the following: public policies, macroeconomic context, geographic characteristics, road and telecommunications infrastructure, among others (Sotomayor *et al.* 2018).

credit. They indicate that the mother tongue is really a *proxy* for distance and location in rural areas far from towns and markets, since people whose mother tongue is not Spanish tend to live farther from developed markets, where there is a financial presence (Alvarado and Pintado 2017). It is therefore suggested that distance from markets is a relevant variable in determining access to credit.

Although the previous study highlights the relevance of distance in the Peruvian context, it does not analyze the positive externalities that could result from proximity to other towns with direct access to the financial system. However, De la Riva Agüero (2009) studied a sample of rural producers in Cusco and Puno who participated in a poverty reduction program to analyze whether transport costs to access developed markets to sell their products affected the accumulation of long-term assets and durable goods. He finds that greater distance is associated with a higher level of investment in physical assets. This suggests that more distant producers must invest more to remain competitive in the market and avoid a decline in their households' well-being.

This study shows that there are spatial factors across Peru's regions that directly affect productivity and indirectly affect well-being, especially for rural families. In a complementary vein, Alfageme and Ramírez-Rondán (2016) note that analyses of the determinants of financial service usage focus on household characteristics, but that some variable capturing the supply of financial services should be incorporated. For all these reasons, since not all of the population has direct access to financial services, the distance to districts that do have formal financial service points available within their territories may generate some positive *spillover* effect on the well-being of families in closer physical proximity to them. This study offers a first approach to capturing this effect.

Finally, regarding informal credit, León *et al.* (2016) analyze the financial requirements of Mypes and whether the financial system currently meets them. The authors evaluate the effect of credit use and find key determinants for designing policies that enable the expansion of services for this segment. After controlling for small entrepreneurs' self-exclusion from the financial system, they find that credit use fails to transform their productive capacity, since the loans received have only a small effect on estimated productivity levels. However, the authors acknowledge that these findings are preliminary and were used to contrast the results of qualitative information revealed in interviews with entrepreneurs.

In conclusion, there are encouraging indicators regarding access to the Peruvian financial system, thanks to the recent expansion of service points. However, it is necessary to deepen the analysis of the effect of the dimensions of financial inclusion on the well-being of Peruvian families in rural areas. Of particular interest is the effect of physical proximity to districts with financial system service points on the consumption patterns of lower-income households living in districts without a financial presence. To do so, the regressions must account for the effect that the number of branches, credits or financial loans has had on the consumption decisions of households that are served, in order to capture the appropriate effects. The next chapter therefore analyzes the effect of financial usage on these consumption decisions.

## Chapter III. Analytical framework and methodology

A family's consumption dynamics can change drastically after gaining access to microcredit. Even if microcredit ends up being beneficial in the long run, not all household expenditures should be expected to change positively and significantly in the same direction and at the same time. To make these dynamics easier to understand, Banerjee *et al.* (2015) propose a simple model of individuals' consumption decisions to understand the potentially diffuse impacts of microcredit on the composition of expenditure.

This model is based on the basic premise that potential borrowers have a spending opportunity that generates intertemporal benefits. Furthermore, Banerjee *et al.* (2015) assume that credit is the only channel enabling this investment since, without access to this resource, the cut in current consumption required to finance the durable good or asset with intertemporal benefits may be unacceptably large for individuals.

### 1. Theoretical framework

It is assumed that there are only two periods and that a representative household consumer can spend their money on two types of goods: durable and non-durable. Non-durable goods, denoted by $c_n$, are fully divisible and consumed during the period in which they are purchased. Durable goods, on the other hand, last over time, are indivisible, cost $c_d$ and yield benefits of $a\,c_d$ in each period. Furthermore, accumulating additional durable goods does not yield a greater benefit in this model.

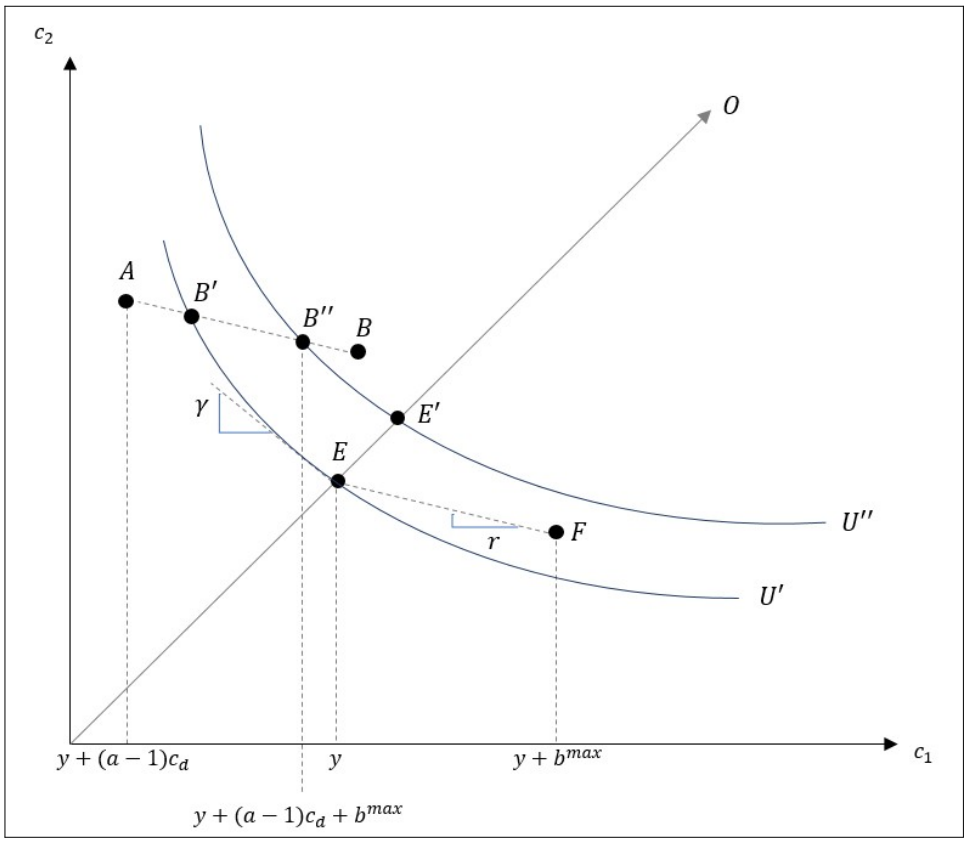
For households, durable and non-durable services are perfect substitutes, so the consumer's per-period utility function is $U(c)$, where $c = c_n$ if the durable good has not been purchased in the current or previous period, and $c = c_n + a\,c_d$ otherwise. Assuming that $0 < a < 1$, the purchase of the durable good entails a net loss in the first period ($(a-1)\,c_d < 0$), but could still have a positive impact, because the consumer discounts the future at a rate '$\gamma$' and maximizes total utility $\left((a-1)\,c_d + \left(\frac{a\,c_d}{1+\gamma}\right) \gtrless 0\right)$.

The representative household individual earns labor income, $y$, in units of the non-durable good in each period, and there is no possibility of saving, so all income is spent in each period. However,

the household has the option to borrow up to an amount $b^{max}$ for one period at a gross interest rate $r$, and cannot take out another loan until it repays its debt. The purchase of the durable good is assumed to cost more than the maximum possible debt: $c_d > b^{max}$.

Under this model proposed by Banerjee *et al.* (2015), households clearly have two non-exclusive options: buy the durable good or take out a loan. Figure 5 plots a pair of indifference curves ($U^{'}, U^{''}$) and possible consumption bundles to facilitate the analysis of consumption options and their effects on utility. The horizontal axis shows consumption in the first period and the vertical axis consumption in the second period.

**Figure 5. Effects of microcredit on intertemporal consumption decisions on household utility under the Banerjee *et al.* (2015) model**




Source: Banerjee *et al.* 2015. Authors' elaboration

Curve $U^{'}$ shows the level of utility consistent with consuming the initial endowment. This bundle $(y, y)$ is denoted by point *E* on the curve. At this point, the slope of the curve is $1/\gamma$. Line *EF*

shows all the consumption options resulting from borrowing in the first period without purchasing the durable good; the slope of this line is determined by the prevailing interest rate ($r$). Under the proposed assumptions, consumers have incentives to access credit.

On the other hand, households have the option to buy a durable good. Point A represents the extreme case in which only this good is purchased without credit financing, which yields a lower level of utility than the initial endowment. However, if a loan is taken, consumption bundles along curve AB become accessible. According to figure 5, households will prefer to borrow and acquire the durable good, since both actions together allow them to increase second-period consumption without suffering a large consumption penalty in the first period. It is important to clarify that a key variable for this statement to hold is the maximum loan amount ($b^{max}$). If the amount is low, it may only be possible to consume a bundle yielding the same utility as the endowment (point $B'$). On the other hand, if the loan amount is high, first-period consumption could fall (point $B''$) or consumption could increase in both periods (point $B$).

Likewise, an increase in the loan interest rate would have a negative effect on household utility, since it reduces second-period consumption because the loan has to be repaid. Thus, in a scenario where $r > 1/\gamma$, there would be no incentive to choose credit on its own, although it would still be rational to buy the durable good with debt, since it could improve the family's well-being depending on the values of the other variables.

Additionally, access to microcredit can create interesting dynamics for households, since it allows consumption to expand in the long run but can also permanently reduce certain types of consumption while increasing others in their place. In other words, a fall in a specific consumption item is not definitive evidence that microcredit has failed to increase the provision of goods for households (Banerjee *et al.* 2015).

Although the model only analyzes two periods, some results can be extrapolated to a situation with infinite periods. First, the benefits of buying durable goods could be perceived over more periods, making the purchase of this type of good even more attractive. Although the model assumes that accumulating several durable goods does not provide a higher level of utility, the existence of maintenance expenditure on these goods or the future replacement of depreciated durable goods should be considered. Also, as debt is repaid, it becomes feasible to take on new loans that allow households to diversify their holdings of durable goods.

This study uses this brief model as a theoretical basis for analyzing total consumption expenditure and its distribution by dividing it into two groups: long-term expenditure (durable goods) and current consumption (non-durable goods) for Peruvian families. There is thus a parallel with the Banerjee *et al.* (2015) model, since long-term investment is considered to be any expenditure on goods or services that fosters the generation of human capital or working capital for household members. It also allows for a time lag between access to credit and the benefits derived from it, a feature that must be taken into account when carrying out the empirical exercise.

Regarding the estimation of the effect of distance to a financial system service point, the model proposed by Burgess and Pande (2005) is taken as the baseline. The authors use a linear OLS model to analyze the impact of the number of financial system access points and the amount of loans and deposits on poverty levels. The baseline function for the empirical analysis is as follows:

$$y_{¿} = \alpha_i + \beta_t + \phi B_{¿}^{R} + \varepsilon_{¿}$$

Where $y_{¿}$denotes a state's poverty rate, $B_{¿}^{R}$ is the cumulative number of access points opened per capita in rural localities with no prior financial presence, and $\alpha_i$ and $\beta_t$ are state and year fixed effects. To ensure the exogeneity of the variable of interest, the authors use the instrumental variables method to estimate $B_{¿}^{R}$. They also use the following control variables: state population density, the log of per capita income, and the log of towns with fewer than 10,000 people.

The premise of this research is similar to that of the present study, since it evaluates the impact of financial system access and usage variables on a societal well-being variable. Unlike Burgess and Pande (2005), this study uses district coordinates to estimate distances and classify rural populations without a financial presence in order to identify whether there is a *spillover* effect of different magnitudes depending on the degree of proximity to districts served by the financial system. It should be noted that those authors acknowledge that they could not perform a more disaggregated analysis with the available data, but stress its relevance for the literature. Furthermore, in the present research, the well-being variable used is linked to household-level consumption rather than district poverty.

## 2. Information sources and data

The study uses two main information sources. The first is the SBS statistical bulletin "Reach and market share" for multiple banking and municipal savings banks, which details the number of loans and deposits by branch, and the distribution of ATMs and banking agents for each public

and private bank at the district level for the 2009 - 2017 period. Additionally, we use the results of the Enaho Household Expenditure module, which details household consumption in the following categories: Food and beverages, Personal care goods and services, Housing services, Housing maintenance, Recreation, Household equipment, Furniture and fixtures, Transport and communications, Transfers, Clothing and footwear, Education, Health, and Other.

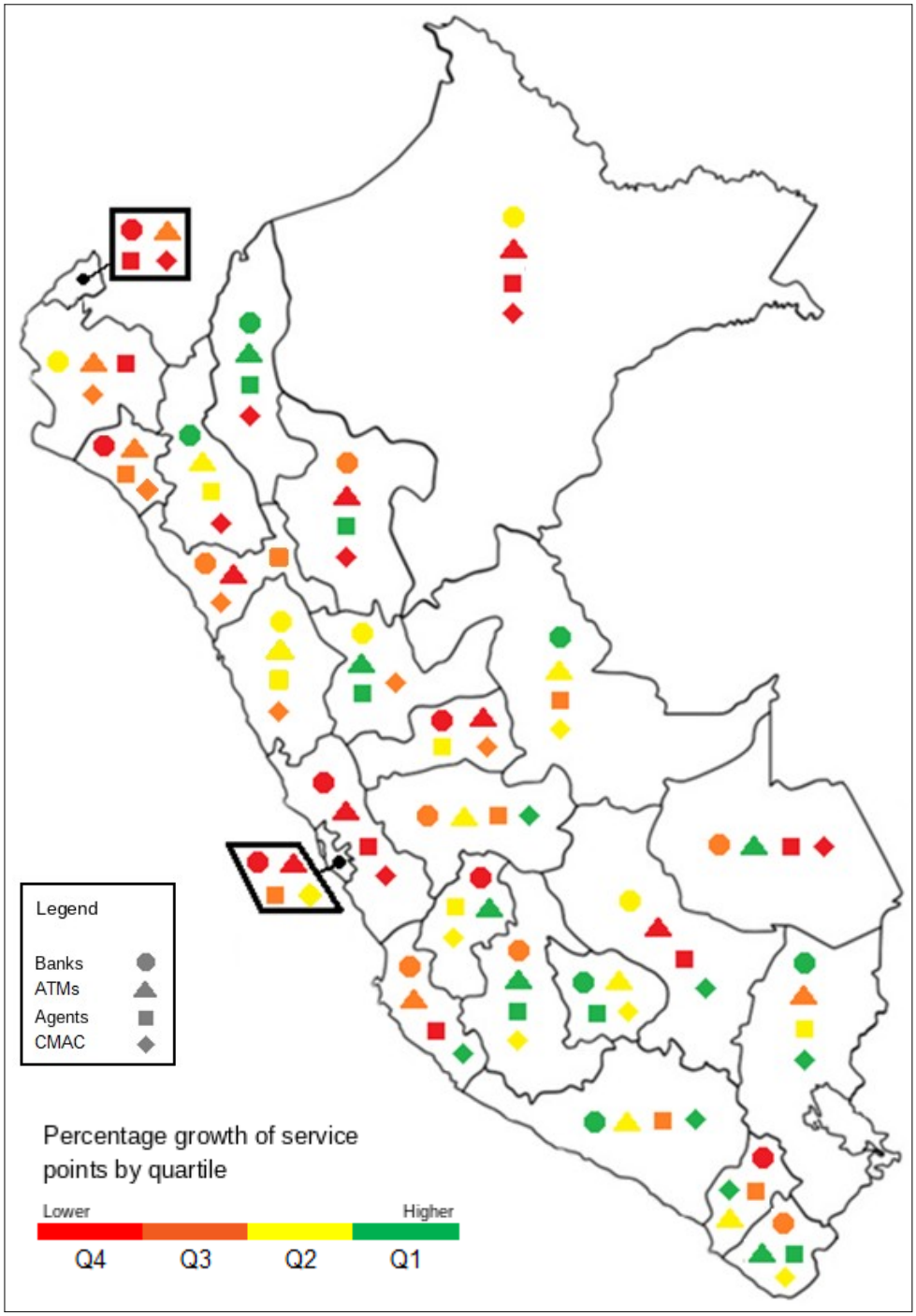
Using the SBS data, various district-level financial inclusion variables were constructed, since this level of territorial disaggregation allows merging with Enaho household data. For the access variables, the number of municipal savings banks, bank branches and banking agents in each district was calculated for each year of the period analyzed[11]. To control for district area, these variables were modified by dividing them by the district's area and multiplying by 100. This yields standardized concentration measures of access to financial services at the district level per 100 $km^2$.

For the financial service usage variables, the number of loans and deposits by district was considered. In the case of loans, the type of financial institution granting them was distinguished. The variables chosen are therefore the total amount of bank loans, the total amount of municipal savings bank loans, and the total amount of deposits. In this case, to control for district density, the district usage variables were estimated per capita.

Since financial system infrastructure may have a lagged effect on investment and consumption decisions, the evolution of these access variables between 2009 and 2017 was analyzed nationwide. During this period, banks increased their supply by 58.78% and CMACs by 116.76%, while ATMs and banking agents increased their national presence by 315.99% and 1,068.79%, respectively.

**Figure 6. Cumulative percentage change in departmental growth of financial system supply by type of access point (2009 - 2017)**

[11] Because of the high correlation between the number of banks and ATMs (over 95%) only the first of these variables was kept in the regressions and the other was omitted.




Source: SBS 2019. Authors' elaboration

Additionally, the percentage change of all these forms of financial supply was analyzed at the departmental level relative to the installed capacity in 2009. Paradoxically, when these relative growth rates are grouped into quartiles, Lima is the only department that always falls in the bottom quartile regardless of the type of financial system supply. Figure 6 shows the results of this

classification for all departments[12].

Finally, using loan data by institution, the Herfindahl-Hirschman Index (HHI) was calculated for each district with a direct financial presence[13]. This market concentration measure was used to segment districts into three categories according to the district HHI range: high, medium and low competition. Three dummy variables were therefore constructed reflecting a district's membership in these segments. These variables were included as a *proxy* for the level of informality since, the greater the competition among formal financial institutions, the more financing options there should be and, therefore, the less need to resort to informal loans. Conversely, in a situation of low competition with few financial institutions, families may have to turn to the informal sector to meet their credit needs (Alvarado *et al.* 2001).

The study's independent variables must make it possible to evaluate the *spillover* effect of the supply of financial service access points on the consumption patterns of families in rural areas in districts without a physical financial presence. To this end, the banking agent network is used to find which districts do not have an access point in their territory but are relatively close to a district that has at least one banking agent.

To identify which districts could experience a *spillover* effect, the distance between a district without a banking agent and the nearest district with one is estimated using the coordinates of every district nationwide. With these data, three dummy variables are generated to classify districts without service points into the following categories according to their distance in kilometers:

- $(0 < Dist \leq 20)$: Districts less than 20 km from another district that has at least one banking agent.
- $(20 < Dist \leq 50)$: Districts more than 20 km but no more than 50 km from another district that has at least one banking agent.
- $(50 < Dist)$: Districts more than 50 km from another district that has at least one banking agent.

The Haversine and Vincenty formulas were used to calculate these distances. According to Mahmoud and Akkari (2016), both formulas are more than 99% accurate in calculating the shortest path between two coordinates. The higher of the two estimates is therefore used as the

[12] The financial supply data underlying this analysis are presented in annex 1.
[13] Both multiple banking institutions and municipal savings banks were considered in calculating the HHI.

baseline distance for the analysis.

Regarding the Enaho data, annual household expenditure was calculated for the 13 consumption categories of the Household Expenditure module mentioned above. However, according to Deaton (2018), although it is necessary to disaggregate consumption categories when collecting data in a large-scale survey in order to obtain appropriate expenditure estimates, analyzing such disaggregated consumption categories could lead to an overestimation of the effects of the independent variables. Total expenditure across categories was therefore divided into only two concepts: long-term expenditure and total current expenditure.

As already mentioned, the concept of long-term expenditure or durable goods in the context of this research encompasses all expenditure on goods or services that fosters the generation of human capital or working capital in households. Therefore, to calculate investment, the following of the 13 consumption categories were aggregated: Housing maintenance, Housing services, Household equipment, Furniture and fixtures, Education, and Health. Based on this calculation, the share of expenditure on long-term durable goods ($\alpha_i^{LP}$) was constructed. Additionally, the effect on total expenditure ($\ln C_i$) is also analyzed, so that with both measures it is possible to see whether there is only a reallocation of expenditure or whether the amount spent also increases.

This study considers data on all families with a household head aged between 14 and 70 in the 2010 - 2017 period who reported their expenditure in the 13 expenditure categories mentioned. Additionally, since the research seeks to identify the effect on rural areas, the sample only includes families living in population centers with fewer than 4,001 dwellings or fewer than 20,000 inhabitants. Furthermore, based on total expenditure, the top and bottom 1% were removed from the sample to eliminate *outliers*. Under these conditions, the final sample includes 104,757 families over 8 years. Table 1 shows the descriptive statistics of the main variables of analysis. In addition, annex 2 presents a detailed description of all the variables used in the empirical exercise.

**Table 1. Descriptive statistics of the relevant variables (Enaho 2010-2017)**

| Variable | Obs. | Mean | Std. Dev. | Min | Max |
|---|---|---|---|---|---|
| Share of long-term expenditure | 104.757 | 0,08553 | 0,079319 | 0 | 0,994976 |
| Total expenditure | 104.757 | 9.556,53 | 10.960,94 | 24 | 708.419,5 |
| Quechua | 104.757 | 0,33656 | 0,472535 | 0 | 1 |
| Other languages | 104.757 | 0,06663 | 0,249382 | 0 | 1 |
| User (1 = financial usage) | 104.757 | 0,663087 | 0,472657 | 0 | 1 |
| Education level | 104.752 | 4,400021 | 2,014323 | 1 | 11 |
| Sex (1 = male) | 104.757 | 0,806848 | 0,394773 | 0 | 1 |
| Married (1 = married) | 104.757 | 0,417662 | 0,493176 | 0 | 1 |

| | | | | | |
|---|---|---|---|---|---|
| Age | 104.757 | 34,41286 | 15,24471 | 14 | 70 |
| Not working (1 = not working) | 104.757 | 0,095535 | 0,293954 | 0 | 1 |
| Self-employed (1 = self-employed) | 104.757 | 0,618412 | 0,485779 | 0 | 1 |
| Employer (1 = employer) | 104.757 | 0,076062 | 0,265098 | 0 | 1 |
| Stratum* | 104.757 | 6,539783 | 1,028784 | 5 | 8 |
| Bank loans $_t$ (per capita) | 104.736 | 0,198834 | 0,982146 | 0 | 19,3437 |
| Caja loans $_t$ (per capita) | 104.736 | 0,207935 | 0,565829 | 0 | 5,997765 |
| Bank deposits $_t$ (per capita) | 104.736 | 0,184539 | 0,974664 | 0 | 17,43399 |
| Caja deposits $_t$ (per capita) | 104.736 | 0,077182 | 0,286533 | 0 | 4,575157 |
| Banks $_t$ (per 100 km$^2$) | 104.736 | 0,166233 | 1,337313 | 0 | 48,19277 |
| Municipal savings banks $_t$ (per 100 km$^2$) | 104.736 | 0,187607 | 1,329972 | 0 | 36,14458 |
| ATMs $_t$ (per 100 km$^2$) | 104.736 | 0,573757 | 5,268928 | 0 | 337,3494 |
| Banking agents $_t$ (per 100 km$^2$) | 104.736 | 3,674529 | 33,15014 | 0 | 2.108,434 |
| High comp. | 104.757 | 0,067169 | 0,250315 | 0 | 1 |
| Medium comp. | 104.757 | 0,048006 | 0,213781 | 0 | 1 |
| Low comp. | 104.757 | 0,126213 | 0,33209 | 0 | 1 |
| $(0 < Dist \leq 20)_t$ | 104.736 | 0,240376 | 0,427314 | 0 | 1 |
| $(20 < Dist \leq 50)_t$ | 104.736 | 0,22053 | 0,414606 | 0 | 1 |
| $(50 < Dist)_t$ | 104.736 | 0,050006 | 0,217959 | 0 | 1 |

*The geographic stratum is a portion of geographic area determined according to the number of dwellings. This variable was used to identify population centers with fewer than 20,000 inhabitants, associated with values 5, 6, 7 and 8.
Source: Authors' elaboration

## 3. Empirical exercise

For the consumption models, annual cross-sectional Enaho data between 2010 and 2017 were used, since the survey's panel modules do not include disaggregated expenditure data. Since there is no information on the same families throughout the period of analysis, Verbeek (2008) recommends grouping families into stable groups with common characteristics that meet conditions of consistency, exogeneity and relevance over time. These models use pseudo-panels that define a new unit of analysis at the cohort level, defined on the basis of observable time-invariant characteristics that generate a complete partition of the analyzed sample, that is, that classify each family into exactly one cohort (Guillerm 2017).

According to Guillerm (2017), selecting the variables to define the cohorts involves a *trade-off* between bias and variance. This is because the size of each cohort must be large enough to limit the possibility of measurement errors in the within-cohort variable means, which are important estimates for the associated methodologies. However, to obtain large cohorts, few variables must be used to delimit the groups, which can produce cohorts with dissimilar characteristics that are subject to greater bias and imprecision in the estimation of the model parameters. On the other hand, using many variables to define the cohorts means that the number of families per group decreases and can therefore produce non-constant cohort fixed effects.

These problems in defining cohorts imply accepting a premise that is difficult to test: that the means of the explanatory variables for the sample cohorts may vary while the means of the population cohorts show no variability at all (Guillerm 2017). However, according to Duong and Nghiem (2014), pseudo-panel models make it possible to overcome some limitations of regular panels, since the data do not need to be adjusted for attrition of individuals, and they achieve greater precision in the calculation of standard errors.

For the baseline model of this research, cohorts were formed using the locality stratum, mother tongue, and birth-year ranges of the household head. These variables have been used in several previous studies to segment cohorts (Verbeek 2008, Duong and Nghiem 2014, Guillerm 2017). With these cohorts it is possible to estimate a fixed effects model, also known as a *within* model, by estimating the mean of the relevant variables for each cohort. For simplicity, the equations listed below use the following reduced version of the *within* variables:

$$\widetilde{X}_{¿} = X_{¿} - \sum_{i=1}^{N} X_{¿}$$

Additionally, since the expansion of financial supply results from market analysis by private firms, it is important to consider the possibility that there are unobserved variables related to both expenditure and financial inclusion measures. For example, the expansion of the service point network may increase residents' business skills, leading families to spend more on durable goods or long-term assets (Berhane and Gardebroek 2011). Instrumental variables are therefore used to argue for the exogeneity of the access and usage variables.

To instrument the variables, the one-period lag of the access and usage variables was used. In this way, present expenditure can be attributed to past financial supply, which is not highly correlated with the unobservable characteristics of the current period (Berhane and Gardebroek 2011). The first stage of the baseline pseudo-panel model is therefore defined by the following equation:

$$\widetilde{Z}_{¿} = \widetilde{X}_{¿}\gamma_i + \delta_1\widetilde{(Caja\ Loans)}_{¿-1} + \delta_2\widetilde{(Bank\ Loans)}_{¿-1} + \delta_3\widetilde{(Total\ Deposits)}_{¿-1}$$
$$y_t = + \delta_4\widetilde{(Banks)}_{¿-1} + \delta_5\widetilde{(Munic.\ Banks)}_{¿-1} + \delta_6\widetilde{(Bank.\ agents)}_{¿-1}$$
$$y_t = + \delta_7\widetilde{(0 < Dist \leq 20)}_{¿} + \delta_8\widetilde{(20 < Dist \leq 50)}_{¿} + \mu_{¿} \dots \text{(*)}$$

Where $Z_{¿}$ is the vector of independent variables predicted by the instrumental variables, composed of: [ $\widetilde{(Caja\ Loans)}_{¿}; \widetilde{(Bank\ Loans)}_{¿}; \widetilde{(Total\ Deposits)}_{¿};$

$\widetilde{(Banks)}_{¿}; \widetilde{(Munic.Banks)}_{¿}; \widetilde{(Bank.agents)}_{¿}; \widetilde{(0<Dist\leq 20)}_{¿}; \widetilde{(20<Dist\leq 50)}_{¿}$ ]. $\widetilde{X}_{¿}$ represents the matrix of control variables for household head characteristics (sex, employment status, marital status, education level, and whether they use any financial service[14]); the lagged indirect access variables are $\widetilde{(0<Dist\leq 20)}_{¿-1}$ and $\widetilde{(20<Dist\leq 50)}_{¿-1}$; the lagged per capita usage variables are $\widetilde{Caja\ Loans}_{¿-1}, \widetilde{Bank\ Loans}_{¿-1}, \widetilde{Total\ Deposits}_{¿-1}$; and the lagged direct access variables per 100 km$^2$ are $\widetilde{Banks}_{¿-1}, \widetilde{Munic.Banks}._{¿-1}, \widetilde{Bank.agents}_{¿-1}$.

With the estimates of the current-period access and usage variables, the second-stage regression of the baseline model is run, defined by the following equation:

$$\widetilde{y}_{¿} = \widetilde{X}_{¿}\gamma_i + \beta_1\widetilde{(Caja\ Loans)}_{¿} + \beta_2\widetilde{(Bank\ Loans)}_{¿} + \beta_3\widetilde{(Total\ Deposits)}_{¿}$$
$$y_t = + \beta_4\widetilde{(Banks)}_{¿} + \beta_5\widetilde{(Munic.Banks)}_{¿} + \beta_6\widetilde{(Bank.agents)}_{¿}$$
$$y_t = + \beta_7\widetilde{(0<Dist\leq 20)}_{¿} + \beta_8\widetilde{(20<Dist\leq 50)}_{¿} + \varepsilon_{¿} \ldots (1)$$

Where $\widetilde{y}_t$ is the household well-being measure analyzed. As mentioned above, two well-being measures were used to capture the total effect on consumption patterns: $\widetilde{\alpha}_{¿}^{LP}$, the share of expenditure on durable goods, and $\widetilde{\ln C}_{¿}$, the amount of total expenditure consumed. It is important to note that, because of the way equation (1) is defined, the results are interpreted relative to the effects found for cohorts that are more than 50 km from another district that has at least one banking agent.

Furthermore, four additional variants of the baseline model are proposed, mainly with changes in the direct access and usage variables, in order to identify some subtleties in the effects and as a robustness check. First, since there may be some degree of substitution between bank and caja loans, the interaction of both variables is added to the baseline model to find whether an increase in one type of credit has a positive or negative effect on the increase in the other. The first variant therefore follows this specification:

$$\widetilde{y}_{¿} = \widetilde{X}_{¿}\gamma_i + \beta_1\widetilde{(Caja\ Loans)}_{¿} + \beta_2\widetilde{(Bank\ Loans)}_{¿} + \beta_3\widetilde{(Total\ Deposits)}_{¿}$$
$$y_t = + \beta_4\left[\widetilde{(Bank\ Loans)}_{¿} * \widetilde{(Caja\ Loans)}_{¿}\right] + \beta_5\widetilde{(Banks)}_{¿} + \beta_6\widetilde{(Munic.Banks)}_{¿}$$
$$y_t = + \beta_7\widetilde{(Bank.agents)}_{¿} + \beta_8\widetilde{(0<Dist\leq 20)}_{¿} + \beta_9\widetilde{(20<Dist\leq 50)}_{¿} + \varepsilon_{¿} \ldots (2)$$

[14] To control for the effect of using a financial service, the variable $Usuario_{¿}$ is used, a dummy variable that takes the value of 1 if the household head has a mortgage, receives interest from a financial product or makes payments via the *web*, and 0 otherwise.

As for deposits, since they do not involve a repayment amount, it is unlikely that there is any degree of substitution by type of institution. Even so, bank and CMAC deposits may have differentiated effects on families. Total deposits are therefore disaggregated according to the following equation:

$$\tilde{y}_{¿} = \tilde{X}_{¿}\gamma_i + \beta_1\widetilde{(Caja\ Loans)}_{¿} + \beta_2\widetilde{(Bank\ Loans)}_{¿} + \beta_3\widetilde{(Caja\ Deposits)}_{¿}$$
$$y_t = + \beta_4\widetilde{(Bank\ Deposits)}_{¿} + \beta_5\widetilde{(Banks)}_{¿} + \beta_6\widetilde{(Munic.\ Banks)}_{¿}$$
$$y_t = + \beta_7\widetilde{(Bank.\ agents)}_{¿} + \beta_8\widetilde{(0 < Dist \leq 20)}_{¿} + \beta_9\widetilde{(20 < Dist \leq 50)}_{¿} + \varepsilon_{¿} \ldots (3)$$

Likewise, the effect of the expansion of the access point network may not be linear. Quadratic financial access variables are therefore added to equation (1) to see whether there is an exponential effect as supply increases in areas with greater installed capacity. This yields the following specification:

$$\tilde{y}_{¿} = \tilde{X}_{¿}\gamma_i + \beta_1\widetilde{(Caja\ Loans)}_{¿} + \beta_2\widetilde{(Bank\ Loans)}_{¿} + \beta_3\widetilde{(Total\ Deposits)}_{¿}$$
$$y_t = + \beta_4\widetilde{(Banks)}_{¿} + \beta_5\widetilde{(Munic.\ Banks)}_{¿} + \beta_6\widetilde{(Bank.\ agents)}_{¿}$$
$$y_t = + \beta_7\widetilde{(Banks^2)}_{¿} + \beta_8\widetilde{(Munic.\ Banks^2)}_{¿} + \beta_9\widetilde{(Bank.\ agents^2)}_{¿}$$
$$y_t = + \beta_{10}\widetilde{(0 < Dist \leq 20)}_{¿} + \beta_{11}\widetilde{(20 < Dist \leq 50)}_{¿} + \varepsilon_{¿} \ldots (4)$$

A competitiveness measure is also incorporated into the model to try to capture the effect of informality, given its predominance in rural populations. Thus, three district-level variables are included that classify the districts of sample families according to relative financial competition. It is important to clarify that competitiveness measures have only been estimated for districts with direct financial access and usage, so the omitted *dummy* is "No competition". This variant therefore follows the formula below:

$$\tilde{y}_{¿} = \tilde{X}_{¿}\gamma_i + \beta_1\widetilde{(Caja\ Loans)}_{¿} + \beta_2\widetilde{(Bank\ Loans)}_{¿} + \beta_3\widetilde{(Total\ Deposits)}_{¿}$$
$$y_t = + \beta_4\widetilde{(Banks)}_{¿} + \beta_5\widetilde{(Munic.\ Banks)}_{¿} + \beta_6\widetilde{(Bank.\ agents)}_{¿}$$
$$y_t = + \beta_7\widetilde{(High\ Comp.)}_{¿} + \beta_8\widetilde{(Medium\ Comp.)}_{¿} + \beta_9\widetilde{(Low\ Comp.)}_{¿}$$
$$y_t = + \beta_{10}\widetilde{(0 < Dist \leq 20)}_{¿} + \beta_{11}\widetilde{(20 < Dist \leq 50)}_{¿} + \varepsilon_{¿} \ldots (5)$$

Finally, since much of the data is at the district level, we chose to run a pair of estimations aggregated at the district level following the baseline model. Cohorts are thus built from the districts of each department, and district means are used for the analysis in equation (6). In this scenario, the unit of analysis is the average family in each district. Since families within a district

show great heterogeneity in their behavior, we also considered a model that additionally defines cohorts using mother tongue. Therefore, the segments in equation (7) distinguish families whose household head's mother tongue is Spanish, Quechua or another language.

## 4. Hypotheses

Given the transformative effect of credit on low-income families, through the expansion of the budget constraint, and its multiplier effect on the economy, the financial development of a district is expected to have a positive effect on the well-being of people living in nearby districts.

In this regard, among families living in a district without a physical financial system presence, a shorter distance to a district with a financial system service point in its territory is expected to be directly related to a larger share of their expenditure on durable goods, but to have no effect on total expenditure. The first effect may be due to the lower cost of accessing districts with more developed markets owing to the presence of the financial system. The null effect on total expenditure, on the other hand, may be associated with higher connectivity spending due to remoteness being offset by lower spending on durable goods. Both ideas are tested and explored further in the following sections.

## Chapter IV. Analysis of results

To organize the presentation of the results, we describe how each relevant variable affects both expenditure measures in the *within* models. These results are shown in table 2 for the share of expenditure on durable goods and in table 3 for total expenditure. Furthermore, for the analysis it should be borne in mind that the financial system access variables can be divided into two groups: the number of financial institutions in the district (direct access) and, where there are no financial institutions, the distance to the nearest district with a financial institution (indirect access).

This section mainly contrasts the results for the indirect access variables. However, it is also of interest to compare the effects on families not served by the financial system with those on families living in districts with direct financial access and usage. Finally, the signs and significance of the control variables are outlined in general terms to validate the effects reported in the literature for the Peruvian context.

Regarding indirect financial access, measured by the distance variables, a physical distance of less than 50 km to the nearest district with a banking agent was found to have a positive effect, significant at 99%, on the share of expenditure allocated to durable goods, regardless of the model evaluated. This effect increases as the distance shrinks, as reflected in the coefficient for districts less than 20 km away (between 0.0033 and 0.0041) being almost double that for living between 20 km and 50 km (between 0.0018 and 0.0025) from the nearest district with a financial presence. Thus, greater proximity to the financial system increases the share of expenditure on durable goods among families living in districts without financial supply, in contrast to families living more than 50 km away.

In contrast, the impact on total household expenditure is the opposite. The results show negative coefficients, significant at 99%, for a physical distance of less than 50 km to the nearest district with a banking agent on the total expenditure measure. However, as the distance shrinks, total expenditure decreases less. Thus, the magnitude of the negative effect on total expenditure of being less than 20 km away (between -0.139 and -0.132) is smaller in absolute value than that of living between 20 and 50 km away (between -0.188 and -0.181).

**Table 2. *Within* model and variants: Effect of access to and use of financial services on the share of expenditure on investment**

| Effects on the share of expenditure allocated to investment | | | | | |
|---|---|---|---|---|---|
| | (1) | (2) | (3) | (4) | (5) |
| **Variables** | **Baseline model** | **Loan interaction** | **Disaggr. deposits** | **Squared access** | **HHI** |
| Banks $_t$ | -0,000772 | -0,000663 | -0,000739 | -0,000697 | -0,000745 |
| | (0,000535) | (0,000525) | (0,000537) | (0,000820) | (0,000533) |
| Municipal savings banks $_t$ | 0,000913* | 0,000869 | 0,000922* | 0,000811 | 0,000907* |
| | (0,000534) | (0,000531) | (0,000533) | (0,000922) | (0,000537) |
| Banking agents $_t$ | -1,55e-06 | -2,90e-06 | -9,58e-07 | -1,60e-05 | -7,73e-07 |
| | (1,15e-05) | (1,14e-05) | (1,15e-05) | (2,12e-05) | (1,15e-05) |
| $(0 < Dist \leq 20)_t$ | 0,00334*** | 0,00319*** | 0,00343*** | 0,00330*** | 0,00410*** |
| | (0,000658) | (0,000659) | (0,000658) | (0,000659) | (0,000689) |
| $(20 < Dist \leq 50)_t$ | 0,00181*** | 0,00166** | 0,00190*** | 0,00177** | 0,00258*** |
| | (0,000689) | (0,000689) | (0,000688) | (0,000689) | (0,000718) |
| Caja loans $_t$ | 0,00291*** | 0,00279*** | 0,00394*** | 0,00296*** | 0,00260*** |
| | (0,000619) | (0,000622) | (0,000770) | (0,000676) | (0,000754) |
| Bank loans $_t$ | -0,00159*** | -0,00247*** | -0,00128*** | -0,00156*** | -0,00122** |
| | (0,000499) | (0,000569) | (0,000494) | (0,000525) | (0,000533) |
| Total deposits $_t$ | 0,00262*** | 0,00264*** | | 0,00262*** | 0,00241*** |
| | (0,000415) | (0,000415) | | (0,000415) | (0,000416) |
| (Caja loans $_t$) * (Bank loans $_t$) | | 0,000575 | | | |
| | | (0,000356) | | | |
| Bank deposits $_t$ | | | 0,00284*** | | |
| | | | (0,000445) | | |
| Caja deposits $_t$ | | | -0,000979 | | |
| | | | (0,00186) | | |
| (Banks $_t$)$^2$ | | | | 1,61e-06 | |
| | | | | (2,12e-05) | |
| (Municipal savings banks $_t$)$^2$ | | | | 5,17e-06 | |
| | | | | (2,88e-05) | |
| (Banking agents $_t$)$^2$ | | | | 8,10e-09 | |
| | | | | (1,21e-08) | |
| High comp. | | | | | -0,000446 |
| | | | | | (0,00152) |
| Medium comp. | | | | | 0,00167 |
| | | | | | (0,00151) |
| Low comp. | | | | | 0,00331*** |
| | | | | | (0,000878) |
| Constant | -8,88e-06 | -0,000135 | -8,89e-06 | -2,89e-05 | -9,13e-06 |
| | (0,000241) | (0,000250) | (0,000241) | (0,000242) | (0,000241) |
| Controls? | Yes | Yes | Yes | Yes | Yes |
| Observations | 103.098 | 103.098 | 103.098 | 103.098 | 103.098 |

Robust standard errors in parentheses

*** p<0,01, ** p<0,05, * p<0,1

Source: Authors' elaboration

**Table 3. *Within* model and variants: Effect of access to and use of financial services on total expenditure**

| Effects on total expenditure | | | | | |
|---|---|---|---|---|---|
| | (1) | (2) | (3) | (4) | (5) |
| **Variables** | **Baseline model** | **Loan interaction** | **Disaggr. deposits** | **Squared access** | **HHI** |
| Banks $_t$ | 0,0238*** | 0,0210*** | 0,0239*** | 0,0812*** | 0,0228*** |
| | (0,00413) | (0,00458) | (0,00457) | (0,00810) | (0,00456) |
| Municipal savings banks $_t$ | -0,0296*** | -0,0285*** | -0,0296*** | -0,151*** | -0,0287*** |
| | (0,00387) | (0,00432) | (0,00432) | (0,00923) | (0,00429) |
| Banking agents $_t$ | 0,00048*** | 0,00051*** | 0,00048*** | 0,00137*** | 0,00046*** |
| | (9,08e-05) | (9,86e-05) | (9,77e-05) | (0,000216) | (9,78e-05) |
| $(0 < Dist \leq 20)_t$ | -0,139*** | -0,135*** | -0,139*** | -0,139*** | -0,132*** |
| | (0,00526) | (0,00624) | (0,00621) | (0,00621) | (0,00667) |
| $(20 < Dist \leq 50)_t$ | -0,188*** | -0,184*** | -0,188*** | -0,188*** | -0,181*** |
| | (0,00565) | (0,00676) | (0,00674) | (0,00674) | (0,00718) |
| Caja loans $_t$ | 0,0871*** | 0,0900*** | 0,0901*** | 0,132*** | 0,0796*** |
| | (0,00461) | (0,00518) | (0,00668) | (0,00567) | (0,00620) |
| Bank loans $_t$ | -0,0364*** | -0,0135*** | -0,0354*** | -0,0489*** | -0,0399*** |
| | (0,00331) | (0,00490) | (0,00395) | (0,00406) | (0,00407) |
| Total deposits $_t$ | 0,0579*** | 0,0573*** | | 0,0581*** | 0,0574*** |
| | (0,00290) | (0,00377) | | (0,00380) | (0,00383) |
| (Caja loans $_t$) * (Bank loans $_t$) | | -0,0150*** (0,00204) | | | |
| Bank deposits $_t$ | | | 0,0586*** (0,00403) | | |
| Caja deposits $_t$ | | | 0,0469*** (0,0148) | | |
| (Banks $_t$)$^2$ | | | | -0,00162*** (0,000213) | |
| (Municipal savings banks $_t$)$^2$ | | | | 0,00391*** (0,000277) | |
| (Banking agents $_t$)$^2$ | | | | -6,4e-07*** (1,04e-07) | |
| High comp. | | | | | 0,0534*** (0,0137) |
| Medium comp. | | | | | -0,0231* (0,0138) |
| Low comp. | | | | | 0,0265*** (0,00878) |
| Constant | -0,236*** | 8,852*** | 8,848*** | 8,845*** | 8,848*** |
| | (0,00193) | (0,00239) | (0,00235) | (0,00237) | (0,00235) |
| Controls? | Yes | Yes | Yes | Yes | Yes |
| Observations | 103.098 | 103.098 | 103.098 | 103.098 | 103.098 |

Robust standard errors in parentheses

*** p<0,01, ** p<0,05, * p<0,1

Source: Authors' elaboration

On the other hand, the effect of the direct access variables on the share of household expenditure on durable goods varies by type of financial institution. In all models, except the one that considers quadratic effects of the direct access variables, only the number of municipal savings banks had a positive effect, significant at 90%. This exception arises because the quadratic effects take significance away from the linear effects. Meanwhile, the effect of the number of banks and banking agents on total expenditure was positive and significant at 99% in all models, whereas the number of municipal savings banks had a negative effect on total expenditure, significant at the same level. In model (4), the effect of the squared number of banks, municipal savings banks and banking agents on expenditure is significant at 99% and opposite in sign to their linear coefficients.

As for the other usage variable, the effect of total deposits on the share of expenditure allocated to durable goods and on total expenditure was positive and significant at 99% in all models that included this variable. In the model that disaggregates deposits by type of financial institution, only the effect of bank deposits was positive and significant at 99% on both well-being measures, while municipal savings bank deposits only maintain a positive effect, significant at 99%, on total expenditure.

Regarding the inclusion of variables associated with financial competition, mixed results were found. Model (5) indicates that only a low level of competition has a positive effect, significant at 99%, on the share of expenditure on durable goods, in contrast to the non-significant coefficients for moderate and high levels of competition. On the other hand, the effects of high and low levels of competition on total expenditure were positive and significant at 99%, while moderate competition had a negative effect, significant at 90%.

For the aggregate models, whose results are detailed in table 4, similar results were found. Regarding the indirect access variables, the effect of being between 0 and 20 km from a district with a financial institution in its territory (between 0.0031 and 0.0034) was positive and significant at 99% on the average share of household expenditure allocated to durable goods for families without financial system service points in their territory. Likewise, being between 20 and 50 km away (between 0.0023 and 0.0027) also had a positive effect, although of smaller magnitude and significance, being significant only at 90% and 95% in models (6) and (7), respectively.

Regarding the effect of distance on total household expenditure, it was negative and significant at

99% in both models. As in the non-aggregated models, as the distance shrinks, total expenditure decreases less. Thus, the magnitude of the negative effect on total expenditure of being less than 20 km away (between -0.0748 and -0.0967) is smaller in absolute value than that of living between 20 and 50 km away (between -0.143 and -0.146).

**Table 4. Aggregate model: Effect of access to and use of financial services on the share of expenditure on investment and on total expenditure**

| | **Effects on the share of expenditure allocated to investment** | **Effects on total expenditure** | | |
|---|---|---|---|---|
| | **(6.1)** | **(7.1)** | **(6.2)** | **(7.2)** |
| **Variables** | **Ubigeo model** | **Ubigeo and language model** | **Ubigeo model** | **Ubigeo and language model** |
| Banks $_t$ | -0,000330 | -0,000849 | 0,00671 | 0,0207* |
| | (0,00136) | (0,00142) | (0,0117) | (0,0118) |
| Municipal savings banks $_t$ | 0,000958 | 0,00140 | -0,0159 | -0,0188** |
| | (0,00128) | (0,00136) | (0,0109) | (0,00922) |
| Banking agents $_t$ | -7,97e-06 | -1,25e-05 | 0,000473** | 0,000206 |
| | (1,33e-05) | (1,51e-05) | (0,000215) | (0,000162) |
| $(0 < Dist \leq 20)_t$ | 0,00307*** | 0,00342*** | -0,0748*** | -0,0967*** |
| | (0,00116) | (0,00121) | (0,0141) | (0,0142) |
| $(20 < Dist \leq 50)_t$ | 0,00234* | 0,00274** | -0,143*** | -0,146*** |
| | (0,00124) | (0,00131) | (0,0159) | (0,0158) |
| Caja loans $_t$ | 0,00115 | -7,65e-07 | 0,0926*** | 0,102*** |
| | (0,00102) | (0,00112) | (0,0140) | (0,0130) |
| Bank loans $_t$ | -0,00104 | -3,39e-05 | -0,0155* | -0,0384*** |
| | (0,000842) | (0,00102) | (0,00927) | (0,0102) |
| Total deposits $_t$ | 0,00201*** | 0,00217*** | 0,0328*** | 0,0512*** |
| | (0,000545) | (0,000620) | (0,00868) | (0,0105) |
| Quechua | 0,00555*** | 0,00504*** | -0,148*** | -0,0802*** |
| | (0,00125) | (0,00117) | (0,0165) | (0,0144) |
| Other languages | 0,000574 | -0,00287 | -0,396*** | -0,269*** |
| | (0,00210) | (0,00189) | (0,0311) | (0,0276) |
| Constant | 0,0713*** | 0,0683*** | 8,703*** | 8,748*** |
| | (0,00596) | (0,00615) | (0,0696) | (0,0682) |
| Controls? | Yes | Yes | Yes | Yes |
| Observations | 4.487 | 5.037 | 4.487 | 5.037 |

Robust standard errors in parentheses

*** p<0,01, ** p<0,05, * p<0,1

Source: Authors' elaboration

On the other hand, none of the direct access variables had a significant effect on the average share of household expenditure allocated to durable goods. Moreover, their effect on total expenditure varied in magnitude, direction and even significance. In model (7), the number of banks had a positive effect, significant at 90%, on average household expenditure, while the number of municipal savings banks had a negative effect, significant at 95%. In model (6), on the other hand,

only the number of banking agents had a significant effect on household expenditure, at the 95% level.

Regarding the effect of the usage variables on the share of household expenditure allocated to durable goods, loans were not significant and only the amount of deposits had a positive effect, significant at 99%. By contrast, the effect of loans on total household expenditure was significant in both models. In both models, the effect of total deposits and of municipal savings bank loans on household expenditure was positive and significant at 99%. Conversely, the effect of bank loans on household expenditure was negative and significant at 90% and 95% in models (6) and (7), respectively.

Finally, regarding the control variables, a higher level of education, being married, being a financial system user and being an employer were found to have a positive impact on both the share of expenditure on durable goods and total expenditure. On the other hand, the age of the household head has a significant negative effect on both well-being measures. In addition, if the household head is self-employed, a positive effect is observed on the share of expenditure on non-durable goods, but a negative one on household expenditure. Lastly, being male only had a significant positive effect on total household expenditure.

## Chapter V. Discussion

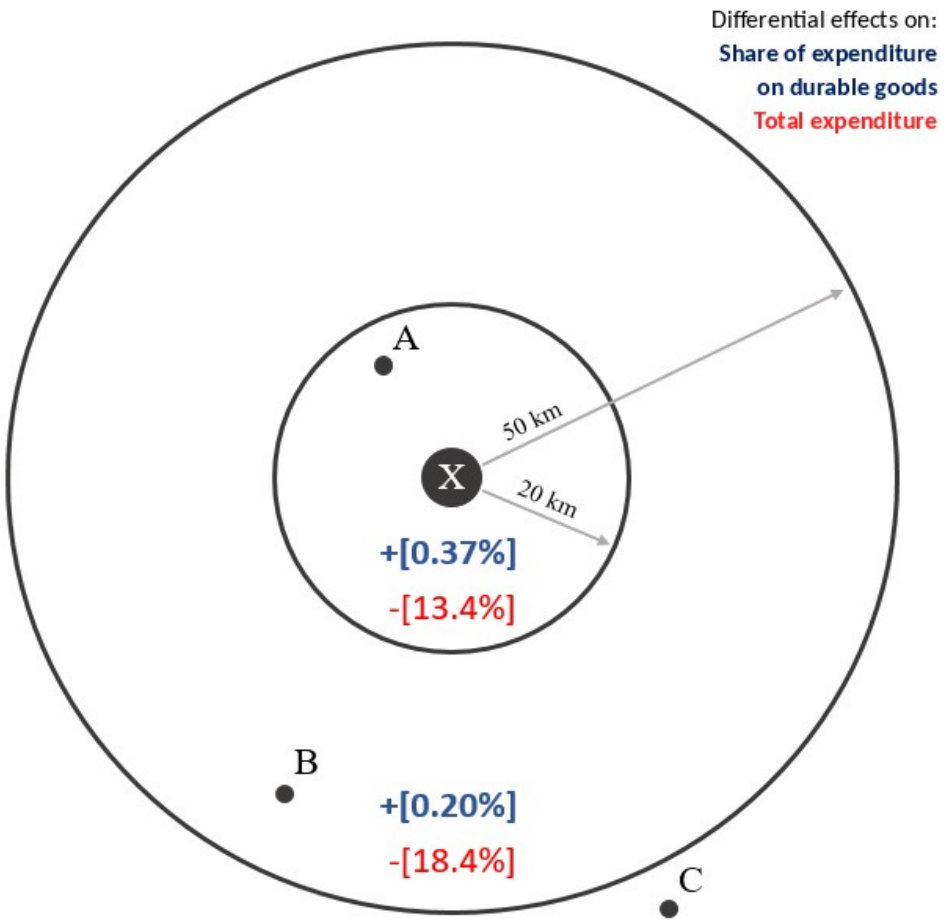
To interpret the above results, we identified two channels through which financial inclusion affects the consumption decisions of families in low population density areas. First, following the model of Banerjee *et al.* (2015), financial products allow families to acquire assets and durable goods that they could not obtain because of their budget constraints. The use of this financial service enables them to reach higher levels of utility. In addition, financial services allow micro-entrepreneurs to develop new businesses, thereby energizing the economy.

Second, a larger number of financial institutions within the same district allows greater spatial coverage in the district, assuming that service points are evenly distributed across the territory. District development is therefore not concentrated at a single point. Thus, regardless of the amount of credit or loans in the districts, greater direct access in itself can energize the economy.

In this way, families close to these developed markets with a financial presence, but lacking a physical financial supply of their own, could be affected indirectly and positively because of their proximity. In this case, there is great variability in the connectivity costs for families living in districts without financial services. The segmentation of this population by distance yields important findings about the differences between these groups of families. To interpret the coefficients, it should be recalled that the base population of the regressions consists of families located more than 50 kilometers from the nearest district with at least one financial system service point.

Accordingly, figure 7 shows a district X that has a banking agent within its territory. It also shows 3 families living in other districts without a physical financial presence in their territories. For all these districts, district X is the nearest one with a financial system service point. Family A is less than 20 km away, family B is more than 20 km and less than 50 km away, and family C is more than 50 km away.

**Figure 7. Analysis of the distance to a financial service point for rural families living in a district without direct financial access**




Source: Authors' elaboration

The results suggest that financial coverage in district X has a *spillover* effect on neighboring districts, with the *spillover* effect understood as the impact on the well-being measures of families living in districts not served by the financial system.

The positive effect of both indirect access variables on the share of expenditure allocated to durable goods would imply that families A and B, which are closer to a district served by the financial system, allocate a relatively larger share of their income to purchasing durable goods or assets/services with future returns. Moreover, family A allocates a larger share of expenditure to this type of good than family B, since its physical distance to district X is shorter. Specifically, using the average results of the *within* models, family B spends 0.20% more than family C on

durable goods, while family A spends 0.37% more. Thus, in relative terms, family A's differential spending is almost double that of family B.

However, a negative effect of indirect access is observed on the total expenditure of families A and B relative to that of family C. Specifically, according to the average results of the *within* models, family B spends 18.4% less than family C on durable goods, while family A spends 13.4% less[15]. Thus, in relative terms, family B's differential spending in absolute value is greater than family A's. Family A therefore has a higher level of expenditure than family B.

Additionally, most direct access and usage variables show a positive effect on total expenditure. This result therefore implies that families living in a district physically unserved by the financial system show lower expenditure than those with direct access. However, among these unserved districts, those farthest from the financial system (family C in the example) incur higher expenditure.

Therefore, families located more than 50 km away spend more than those less than 50 km away, but probably on non-durable goods. This finding suggests that greater distance from the financial system may be related to a subsistence-oriented allocation of resources. Thus, the higher spending may be on transport, since longer distances must be traveled to reach developed markets, or on food, because these families do not benefit from the lower prices resulting from economies of scale or mass production. This interpretation is in line with the argument put forward by De la Riva Agüero (2009), who argued that more distant producers had to spend more to remain competitive. In this case, however, what is found is not a higher level of investment but higher spending on non-durable goods.

Under the argument that greater distance implies high costs of connecting to markets, the results showing lower spending in the closer unserved districts make sense. On the other hand, proximity to markets gives families more options of products to consume, which encourages greater spending. That is why families less than 20 km away were found to spend more than those between 20 km and 50 km away.

Although the level of expenditure is a well-being measure widely used in the literature, spending S/ 500 a month on transport instead of S/ 100 does not necessarily imply a higher level of well-being for families. Incurring greater expenditure to receive the same service is a sign of disparity

[15] Since the model uses the logarithm of total expenditure, the coefficients are semi-elasticities.

in service and coverage, not of a better quality of life. Therefore, in this research, the share of expenditure on durable goods is a more appropriate variable for measuring well-being, since it identifies how much of the expenditure is really productive. This attribute is fundamental in the context of low-income populations, because it makes it possible to identify whether living conditions have improved or whether families are merely subsisting.

Although the focus of this study is on the impact of physical distance to the financial system on well-being, it is important to comment on the direct access and financial usage variables. First, the usage variables were constructed with district-level data, without being able to distinguish how much each family uses. The interpretation therefore refers to the district's financial concentration. Thus, since in most districts there are loans from both types of financial institutions, the signs of the results show that caja loans have had a greater impact than bank loans on the well-being measures. Loans in general have a positive impact, but there is substitution between loans from different institutions.

As for deposits, the positive effect on both expenditure measures shows that this variable is a good indicator of financial development. A larger amount of deposits makes it possible to multiply wealth through rural families' receipt of remittances, bill payments, financial transfers, among others. Additionally, the accumulation of deposits generates funds for financial institutions, which in turn finance productive expenditure and investment (Allen *et al.* 2015). In this way, the economy is energized and household expenditure increases.

As mentioned above, the growth of financial system service points during the period of analysis has been exponential. This has allowed a greater presence in previously unserved districts and expanded coverage within several districts. This decentralization of financial services, captured by the direct access variables, had a positive impact on energizing the economy through increased expenditure. Likewise, the negative sign for the number of municipal savings banks does not necessarily indicate that their presence slows the economy and reduces expenditure. What may be happening is that it is cannibalizing the effect of the other direct access variables, since the expansion of financial supply has been shown to be driven mainly by banking agents. The same phenomenon occurs with the use of credit.

Additionally, the negative and significant effect of the squared number of financial institutions on total expenditure helps explain the effect of intra-district coverage. Since the direct access variables were constructed at the district level, there may be fewer and fewer new areas to cover within districts. The effect on energizing the economy is therefore increasingly smaller. Thus,

there are indications that the expansion of financial supply could have a diminishing marginal effect within already served districts, but a positive effect on unserved ones.

Although the competitiveness index was used to include the level of informality of the financial system in the model, the results indicate that the competition variables have not worked as a good proxy for the degree of financial informality. The literature indicates that places with low competitiveness would have a higher degree of financial informality, which is associated with a larger share of loans for current expenditure (Alvarado *et al.* 2001). Low competitiveness would therefore be expected to have a positive effect on expenditure through greater informality. However, the effect of low competition, although positive, is smaller than that of high competition on household expenditure. Finding a variable that accounts for the important effect of informality therefore remains pending.

## 1. New lines of research

As has been observed, data availability has not allowed clearer and more robust effects of financial inclusion on expenditure patterns to be obtained. In this regard, identifying the effect on both users and non-users, even in nearby physically unserved districts, remains a task for new lines of research. Furthermore, being able to identify whether a district's loans and deposits belong to its residents or to residents of nearby districts would make it possible to observe the *spillover* effect and its channels with more precise information.

In another line of research, identifying whether people borrow informally would make it possible to determine the effect of each of these types of loans on household well-being. Likewise, if the presence of informality is considered as an incumbent firm and the possible entry of a bank as the entrant firm, the costs and difficulties could be analyzed within the framework of a Stackelberg model.

Regarding the results obtained for the indirect access variables, the analysis could be deepened with more precise information on the cost of accessing the financial system. In the case of the direct access variables, it would be ideal to have the time it takes to reach the financial institution from home, the limitations people face in using it, and additional costs. The ENDSF collects some of these data, but not for a large number of periods. In this regard, if instead of the number of branches in a district their locations were available, an indicator of the percentage of a district covered could be obtained. Additionally, with information on population density within a district,

an indicator of the percentage of the population covered could be constructed.

On the other hand, the distance between districts was used to construct the indirect access variables and measure the *spillover* effect. However, new lines of research could consider using other variables, for example, geographic features, available transport, and other factors that affect the cost of access, whether in monetary terms or in time.

## 2. Policy recommendations

Finally, the findings on the importance of distance to the financial system for the type of consumption of rural families, together with the fact that most districts nationwide still lack financial services in their territories, create the need to propose measures that can help improve the current situation. The following public policy recommendations are therefore proposed, in line with the Action Plan of the National Financial Inclusion Strategy (Ministry of Economy and Finance [MEF] 2015).

The results show that the *spillover* effect loses strength beyond 50 km. In this regard, a public policy that promotes and provides incentives for the creation of banking agents in centrally located unserved districts would reduce the gap in access to the financial system not only in that district but also in the surrounding ones. Based on SBS financial access data (2019), most districts isolated from the financial system (more than 50 km from the nearest district with at least one banking agent) were found to be in Loreto and Huancavelica, which accounted for 38.16% and 23.68%, respectively, of the total of 76 unserved districts at the end of 2017. It would therefore be advisable to focus attention on these districts, since they have a greater need for at least one banking agent in their vicinity.

A complementary strategy could be to strengthen financial services in the districts that are the nearest ones for a significant number of unserved districts, since these service points can meet the financial needs of their own residents and of families from other districts. Thus, based on SBS financial access data (2019) at the end of 2017, table 5 identifies the districts that, because of their location and physical financial presence, could have the largest number of neighboring districts dependent on the financial services offered there.

**Table 5. Districts with a physical financial presence that at least 20 unserved districts perceive as the nearest district with a financial system service point**

| District | Province | Department | No. of nearby districts without access to the financial system |
|---|---|---|---|
| Chalhuanca | Aymaes | Apurímac | 25 |
| Zúñiga | Cañete | Lima | 24 |
| Huarochirí | Huarochirí | Lima | 24 |
| Chiquián | Bolognesi | Áncash | 23 |
| Vilcashuamán | Vilcashuamán | Ayacucho | 21 |
| Cotahuasi | La Unión | Arequipa | 20 |

Source: SBS 2019. Authors' elaboration

De Vreyer *et al.* (2009) note that there are areas within Peru that could be classified as spatial poverty traps. The inhabitants of these territories fail to develop because of their lack of access to services and their socioeconomic characteristics, not because of the geographic characteristics of the area. Therefore, strengthening financial services could indeed be beneficial insofar as it improves service endowments and schooling levels, leading to a subsequent impact on consumption.

Likewise, greater public spending on road infrastructure and internet access would reduce access costs for people living in districts without access to the financial system. In this regard, the *spillover* effect of financial services in a district on its neighboring districts would have a greater reach thanks to better connectivity. Moreover, greater infrastructure would reduce transport costs for other basic necessities. This would lower the prices of these goods, allowing families to achieve greater well-being by increasing their consumption without having to spend more.

Finally, another complementary measure would be to promote financial education in rural areas with the support of financial institutions. In this way, the benefits of accessing the financial system could be disseminated, together with education on basic financial concepts. This measure could even help mitigate the popularity of informal credit.

**Conclusions**

In Peru, the financial inclusion literature has been moderately explored, with an emphasis on identifying the determinants of financial access and usage. This study, however, sought to understand how indirect financial access affected the expenditure patterns of rural families, as well as the channels through which it affects economic dynamism.

Regarding *spillover* effects, living closer to a district with a service point was found to allow families without a physical financial presence to spend less and allocate a larger share of their expenditure to

durable goods. This lower spending results from two opposing forces. A shorter distance implies lower costs of connecting to markets, but proximity to markets gives families more spending options. Therefore, families more than 50 km away have higher expenditure than closer families, but families less than 20 km away spend more than those between 20 km and 50 km away.

These results show that greater connectivity between districts is needed in Peru, since 64.9% of districts had no physical financial system presence at the end of 2017. Therefore, the installation of financial institutions in centrally located unserved districts should be encouraged and the most isolated services strengthened. This will increase the share of expenditure on durable goods and reduce connectivity costs, allowing families to spend on other goods.

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**Annexes**

## Annex 1. Annual departmental evolution of financial system supply from multiple banking institutions and municipal savings banks (2009 – 2017)

Annual departmental evolution of banks (2009-2017)

| Department | 2009 | 2010 | 2011 | 2012 | 2013 | 2014 | 2015 | 2016 | 2017 |
|---|---|---|---|---|---|---|---|---|---|
| Amazonas | 2 | 2 | 2 | 2 | 3 | 3 | 5 | 8 | 8 |
| Áncash | 24 | 27 | 28 | 28 | 36 | 38 | 39 | 56 | 54 |
| Apurímac | 3 | 4 | 4 | 4 | 4 | 4 | 6 | 10 | 11 |
| Arequipa | 40 | 48 | 56 | 66 | 72 | 73 | 75 | 99 | 97 |
| Ayacucho | 5 | 6 | 5 | 5 | 6 | 6 | 6 | 9 | 10 |
| Cajamarca | 17 | 20 | 21 | 20 | 23 | 24 | 27 | 48 | 47 |
| Callao | 71 | 79 | 82 | 81 | 82 | 84 | 86 | 87 | 85 |
| Cusco | 24 | 27 | 28 | 32 | 32 | 32 | 40 | 56 | 56 |
| Huancavelica | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 |
| Huánuco | 8 | 10 | 10 | 10 | 10 | 16 | 16 | 18 | 18 |
| Ica | 35 | 42 | 42 | 44 | 48 | 53 | 58 | 62 | 62 |
| Junín | 34 | 42 | 40 | 37 | 43 | 46 | 47 | 53 | 58 |
| La Libertad | 59 | 68 | 68 | 67 | 83 | 84 | 87 | 107 | 105 |
| Lambayeque | 40 | 43 | 39 | 43 | 51 | 53 | 56 | 69 | 64 |
| Lima | 841 | 886 | 919 | 954 | 1.059 | 1.077 | 1.115 | 1.180 | 1.167 |
| Loreto | 12 | 15 | 15 | 16 | 20 | 21 | 23 | 24 | 25 |
| Madre de Dios | 3 | 4 | 4 | 4 | 5 | 5 | 6 | 6 | 6 |
| Moquegua | 11 | 12 | 12 | 11 | 11 | 11 | 12 | 16 | 16 |
| Pasco | 5 | 5 | 6 | 5 | 5 | 4 | 5 | 6 | 6 |
| Piura | 44 | 52 | 55 | 62 | 68 | 71 | 77 | 102 | 103 |
| Puno | 15 | 17 | 18 | 21 | 25 | 24 | 24 | 44 | 43 |
| San Martín | 15 | 18 | 20 | 18 | 20 | 23 | 23 | 28 | 30 |
| Tacna | 14 | 18 | 20 | 21 | 22 | 22 | 22 | 26 | 24 |
| Tumbes | 8 | 9 | 9 | 8 | 8 | 8 | 9 | 12 | 11 |
| Ucayali | 8 | 9 | 9 | 9 | 11 | 14 | 18 | 19 | 19 |
| Total | 1.339 | 1.464 | 1.513 | 1.569 | 1.748 | 1.797 | 1.883 | 2.146 | 2.126 |

Source: SBS 2019. Authors' elaboration

Annual departmental evolution of ATMs (2009-2017)

| Department | 2009 | 2010 | 2011 | 2012 | 2013 | 2014 | 2015 | 2016 | 2017 |
|---|---|---|---|---|---|---|---|---|---|
| Amazonas | 2 | 10 | 2 | 3 | 2 | 4 | 10 | 22 | 22 |
| Áncash | 51 | 74 | 68 | 73 | 20 | 132 | 174 | 299 | 299 |
| Apurímac | 6 | 5 | 5 | 7 | 4 | 11 | 22 | 34 | 34 |
| Arequipa | 127 | 166 | 157 | 238 | 47 | 302 | 421 | 707 | 677 |
| Ayacucho | 15 | 33 | 16 | 17 | 5 | 30 | 77 | 137 | 142 |
| Cajamarca | 35 | 40 | 46 | 55 | 15 | 86 | 134 | 210 | 203 |
| Callao | 165 | 180 | 198 | 222 | 28 | 298 | 421 | 594 | 582 |
| Cusco | 105 | 110 | 120 | 137 | 30 | 182 | 288 | 414 | 372 |
| Huancavelica | 2 | 37 | 7 | 3 | 4 | 6 | 7 | 75 | 77 |
| Huánuco | 13 | 14 | 17 | 17 | 10 | 52 | 67 | 98 | 95 |
| Ica | 83 | 102 | 115 | 141 | 33 | 218 | 351 | 508 | 470 |
| Junín | 62 | 86 | 81 | 96 | 36 | 151 | 244 | 362 | 353 |
| La Libertad | 148 | 175 | 201 | 235 | 33 | 334 | 509 | 687 | 647 |
| Lambayeque | 93 | 125 | 117 | 152 | 25 | 217 | 332 | 489 | 447 |
| Lima | 2.125 | 2.204 | 2.688 | 3.085 | 288 | 4.313 | 5.984 | 8.243 | 7.981 |
| Loreto | 46 | 48 | 51 | 67 | 15 | 88 | 120 | 155 | 150 |
| Madre de Dios | 4 | 6 | 9 | 11 | 6 | 18 | 26 | 38 | 37 |

| Department | 2009 | 2010 | 2011 | 2012 | 2013 | 2014 | 2015 | 2016 | 2017 |
|---|---|---|---|---|---|---|---|---|---|
| Moquegua | 12 | 12 | 15 | 17 | 10 | 24 | 46 | 71 | 69 |
| Pasco | 10 | 8 | 11 | 10 | 9 | 17 | 21 | 35 | 34 |
| Piura | 109 | 115 | 133 | 198 | 39 | 258 | 387 | 561 | 557 |
| Puno | 30 | 27 | 29 | 47 | 10 | 62 | 107 | 148 | 138 |
| San Martín | 25 | 23 | 29 | 30 | 16 | 63 | 80 | 114 | 112 |
| Tacna | 28 | 35 | 40 | 52 | 9 | 69 | 113 | 167 | 172 |
| Tumbes | 13 | 12 | 16 | 20 | 6 | 26 | 41 | 56 | 63 |
| Ucayali | 19 | 21 | 28 | 36 | 9 | 48 | 87 | 122 | 111 |
| Total | 3.328 | 3.668 | 4.199 | 4.969 | 709 | 7.009 | 10.069 | 14.346 | 13.844 |

Source: SBS 2019. Authors' elaboration

Annual departmental evolution of banking agents (2009-2017)

| Department | 2009 | 2010 | 2011 | 2012 | 2013 | 2014 | 2015 | 2016 | 2017 |
|---|---|---|---|---|---|---|---|---|---|
| Amazonas | 10 | 21 | 36 | 31 | 58 | 60 | 115 | 292 | 261 |
| Áncash | 79 | 107 | 139 | 176 | 217 | 342 | 667 | 1.278 | 1.386 |
| Apurímac | 8 | 18 | 24 | 29 | 51 | 81 | 108 | 284 | 357 |
| Arequipa | 285 | 502 | 811 | 962 | 1.329 | 1.900 | 2.470 | 4.554 | 4.605 |
| Ayacucho | 14 | 30 | 48 | 51 | 78 | 120 | 230 | 428 | 403 |
| Cajamarca | 62 | 91 | 162 | 202 | 264 | 421 | 763 | 1.333 | 1.329 |
| Callao | 165 | 245 | 359 | 474 | 703 | 849 | 1.315 | 2.565 | 2.373 |
| Cusco | 107 | 162 | 242 | 298 | 378 | 510 | 764 | 1.032 | 1.341 |
| Huancavelica | 8 | 8 | 25 | 19 | 23 | 35 | 34 | 106 | 135 |
| Huánuco | 21 | 27 | 44 | 64 | 65 | 145 | 278 | 514 | 575 |
| Ica | 116 | 177 | 247 | 293 | 357 | 475 | 677 | 1.493 | 1.384 |
| Junín | 148 | 192 | 329 | 388 | 537 | 734 | 1.057 | 1.931 | 2.164 |
| La Libertad | 219 | 283 | 343 | 445 | 671 | 912 | 1.685 | 3.021 | 3.048 |
| Lambayeque | 162 | 223 | 294 | 385 | 499 | 789 | 1.218 | 2.341 | 2.293 |
| Lima | 2.469 | 3.010 | 4.584 | 5.913 | 8.212 | 9.833 | 13.418 | 25.403 | 22.454 |
| Loreto | 27 | 43 | 49 | 53 | 62 | 78 | 181 | 320 | 365 |
| Madre de Dios | 6 | 9 | 13 | 17 | 14 | 24 | 18 | 59 | 74 |
| Moquegua | 18 | 38 | 56 | 65 | 66 | 117 | 217 | 322 | 283 |
| Pasco | 15 | 23 | 22 | 41 | 58 | 74 | 112 | 189 | 257 |
| Piura | 169 | 203 | 293 | 359 | 461 | 571 | 826 | 1.683 | 1.582 |
| Puno | 37 | 45 | 75 | 96 | 130 | 146 | 357 | 639 | 665 |
| San Martín | 42 | 73 | 124 | 182 | 227 | 340 | 591 | 1.043 | 1.150 |
| Tacna | 30 | 60 | 75 | 97 | 151 | 189 | 432 | 683 | 711 |
| Tumbes | 29 | 36 | 47 | 45 | 56 | 68 | 88 | 169 | 242 |
| Ucayali | 51 | 54 | 73 | 97 | 173 | 262 | 508 | 881 | 786 |
| Total | 4.297 | 5.680 | 8.514 | 10.782 | 14.840 | 19.075 | 28.129 | 52.563 | 50.223 |

Source: SBS 2019. Authors' elaboration

Annual departmental evolution of municipal savings banks (2009-2017)

| Department | 2009 | 2010 | 2011 | 2012 | 2013 | 2014 | 2015 | 2016 | 2017 |
|---|---|---|---|---|---|---|---|---|---|
| Amazonas | 7 | 9 | 9 | 10 | 8 | 11 | 13 | 12 | 12 |
| Áncash | 12 | 15 | 18 | 18 | 19 | 17 | 17 | 20 | 21 |
| Apurímac | 7 | 13 | 14 | 14 | 15 | 17 | 18 | 18 | 19 |
| Arequipa | 16 | 28 | 37 | 43 | 46 | 51 | 57 | 57 | 57 |
| Ayacucho | 7 | 10 | 11 | 12 | 11 | 12 | 14 | 16 | 16 |
| Cajamarca | 15 | 19 | 19 | 19 | 20 | 24 | 25 | 25 | 26 |
| Callao | 3 | 4 | 4 | 5 | 5 | 5 | 5 | 5 | 6 |

| Department | 2009 | 2010 | 2011 | 2012 | 2013 | 2014 | 2015 | 2016 | 2017 |
|---|---|---|---|---|---|---|---|---|---|
| Cusco | 20 | 26 | 30 | 31 | 35 | 39 | 48 | 54 | 57 |
| Huancavelica | 4 | 4 | 6 | 6 | 7 | 8 | 7 | 9 | 10 |
| Huánuco | 10 | 9 | 11 | 12 | 13 | 13 | 13 | 18 | 19 |
| Ica | 10 | 22 | 22 | 24 | 21 | 23 | 23 | 29 | 33 |
| Junín | 15 | 19 | 26 | 26 | 32 | 37 | 42 | 45 | 53 |
| La Libertad | 21 | 25 | 28 | 29 | 30 | 31 | 34 | 37 | 40 |
| Lambayeque | 13 | 16 | 20 | 21 | 21 | 23 | 23 | 24 | 25 |
| Lima | 82 | 98 | 116 | 124 | 127 | 133 | 134 | 139 | 142 |
| Loreto | 8 | 10 | 10 | 10 | 10 | 11 | 12 | 12 | 13 |
| Madre de Dios | 7 | 9 | 10 | 10 | 11 | 11 | 11 | 10 | 10 |
| Moquegua | 4 | 4 | 5 | 9 | 10 | 10 | 13 | 13 | 14 |
| Pasco | 5 | 5 | 5 | 5 | 8 | 8 | 9 | 9 | 9 |
| Piura | 30 | 40 | 48 | 48 | 51 | 53 | 53 | 54 | 58 |
| Puno | 17 | 28 | 28 | 31 | 37 | 38 | 43 | 47 | 49 |
| San Martín | 15 | 15 | 15 | 15 | 17 | 19 | 18 | 18 | 23 |
| Tacna | 7 | 9 | 12 | 12 | 13 | 14 | 16 | 16 | 17 |
| Tumbes | 6 | 8 | 10 | 10 | 10 | 10 | 10 | 9 | 10 |
| Ucayali | 5 | 5 | 7 | 8 | 10 | 10 | 11 | 11 | 11 |
| Total | 346 | 450 | 521 | 552 | 587 | 628 | 669 | 707 | 750 |

Source: SBS 2019. Authors' elaboration

Cumulative percentage change in departmental growth of financial system supply by type of access point (2009-2017)

| Department | Banks | ATMs | Bank. agents | CMAC |
|---|---|---|---|---|
| Amazonas | 300,00 % | 1000,00 % | 2510,00 % | 71,43 % |
| Áncash | 125,00 % | 486,27 % | 1654,43 % | 75,00 % |
| Apurímac | 266,67 % | 466,67 % | 4362,50 % | 171,43 % |
| Arequipa | 142,50 % | 433,07 % | 1515,79 % | 256,25 % |
| Ayacucho | 100,00 % | 846,67 % | 2778,57 % | 128,57 % |
| Cajamarca | 176,47 % | 480,00 % | 2043,55 % | 73,33 % |
| Callao | 19,72 % | 252,73 % | 1338,18 % | 100,00 % |
| Cusco | 133,33 % | 254,29 % | 1153,27 % | 185,00 % |
| Huancavelica | 0,00 % | 3750,00 % | 1587,50 % | 150,00 % |
| Huánuco | 125,00 % | 630,77 % | 2638,10 % | 90,00 % |
| Ica | 77,14 % | 466,27 % | 1093,10 % | 230,00 % |
| Junín | 70,59 % | 469,35 % | 1362,16 % | 253,33 % |
| La Libertad | 77,97 % | 337,16 % | 1291,78 % | 90,48 % |
| Lambayeque | 60,00 % | 380,65 % | 1315,43 % | 92,31 % |
| Lima | 38,76 % | 275,58 % | 809,44 % | 73,17 % |
| Loreto | 108,33 % | 226,09 % | 1251,85 % | 62,50 % |
| Madre de Dios | 100,00 % | 825,00 % | 1133,33 % | 42,86 % |
| Moquegua | 45,45 % | 475,00 % | 1472,22 % | 250,00 % |
| Pasco | 20,00 % | 240,00 % | 1613,33 % | 80,00 % |
| Piura | 134,09 % | 411,01 % | 836,09 % | 93,33 % |
| Puno | 186,67 % | 360,00 % | 1697,30 % | 188,24 % |
| San Martín | 100,00 % | 348,00 % | 2638,10 % | 53,33 % |
| Tacna | 71,43 % | 514,29 % | 2270,00 % | 142,86 % |
| Tumbes | 37,50 % | 384,62 % | 734,48 % | 66,67 % |
| Ucayali | 137,50 % | 484,21 % | 1441,18 % | 120,00 % |
| Total | 58,78 % | 315,99 % | 1068,79 % | 116,76 % |

Source: SBS 2019. Authors' elaboration

## Annex 2. Description of explanatory variables

Description of the explanatory variables of the empirical exercise

| Variable | Description | Type of variable |
|---|---|---|
| Banks $_t$ | Number of banks in the current period, at the district level, divided by the district's area and multiplied by 100. | Access variable |
| Municipal savings banks $_t$ | Number of municipal savings banks in the current period, at the district level, divided by the district's area and multiplied by 100. | Access variable |
| Banking agents $_t$ | Number of banking agents in the current period, at the district level, divided by the district's area and multiplied by 100. | Access variable |
| $(0 < Dist \leq 20)_t$ | A *dummy* indicating whether the family lives in a district less than 20 km from another district that has at least one banking agent. | Access variable |
| $(20 < Dist \leq 50)_t$ | A *dummy* indicating whether the family lives in a district more than 20 km but no more than 50 km from another district that has at least one banking agent. | Access variable |
| (Banks $_t$)$^2$ | Number of banks in the current period, at the district level, divided by the district's area and multiplied by 100, squared. | Access variable |
| (Municipal savings banks $_t$)$^2$ | Number of municipal savings banks in the current period, at the district level, divided by the district's area and multiplied by 100, squared. | Access variable |
| (Banking agents $_t$)$^2$ | Number of banking agents in the current period, at the district level, divided by the district's area and multiplied by 100, squared. | Access variable |
| Bank loans $_t$ | Total district-level loans provided by banks, divided by the district's population. | Usage variable |
| Caja loans $_t$ | Total district-level loans provided by municipal savings banks, divided by the district's population. | Usage variable |
| Total deposits $_t$ | Total district-level time deposits, divided by the district's population. | Usage variable |
| Bank deposits $_t$ | Total district-level demand deposits provided by banks, divided by the district's population. | Usage variable |
| Caja deposits $_t$ | Total district-level demand deposits provided by municipal savings banks, divided by the district's population. | Usage variable |
| Banks $_{t-1}$ | Number of banks in the previous period, at the district level, divided by the district's area and multiplied by 100 | Instrument |
| Municipal savings banks $_{t-1}$ | Number of municipal savings banks in the previous period, at the district level, divided by the district's area and multiplied by 100 | Instrument |
| Banking agents $_{t-1}$ | Number of banking agents in the previous period, at the district level, divided by the district's area and multiplied by 100 | Instrument |
| Bank loans $_{t-1}$ | Amount of district-level loans provided by banks in the previous period, divided by the district's population. | Instrument |
| Caja loans $_{t-1}$ | Total district-level loans provided by municipal savings banks in the previous period, divided by the district's population. | Instrument |
| Total deposits $_{t-1}$ | Total amount of district-level deposits in the previous period, divided by the district's population. | Instrument |
| User | A *dummy* that takes the value of 1 if the household has taken a loan to finance its house, land or other, and 0 otherwise. | Control variable |
| Education level | Education level of the household head. | Control variable |
| Sex | A *dummy* that takes the value of 1 if the household head is male, and 0 if female. | Control variable |
| Married | A *dummy* that takes the value of 1 if the household head is married, and 0 otherwise. | Control variable |
| Age | Age of the household head. | Control variable |

| Variable | Description | Type of variable |
|---|---|---|
| Not working | A *dummy* that takes the value of 1 if the household head does not work, and 0 otherwise. | Control variable |
| Self-employed | A *dummy* that takes the value of 1 if the household head is self-employed, and 0 otherwise. | Control variable |
| Employer | A *dummy* that takes the value of 1 if the household head is an employer, and 0 otherwise. | Control variable |
| High comp. | A *dummy* that takes the value of 1 if the district where the family lives has a credit HHI between 0.13 and 0.42, and 0 otherwise. | Control variable |
| Medium comp. | A *dummy* that takes the value of 1 if the district where the family lives has a credit HHI between 0.42 and 0.71, and 0 otherwise. | Control variable |
| Low comp. | A *dummy* that takes the value of 1 if the district where the family lives has a credit HHI between 0.71 and 1, and 0 otherwise. | Control variable |
| Quechua | A *dummy* that takes the value of 1 if the household head's mother tongue is Quechua, and 0 otherwise. | Control variable |
| Other languages | A *dummy* that takes the value of 1 if the household head's mother tongue is neither Spanish nor Quechua, and 0 otherwise. | Control variable |

Source: Authors' elaboration

## Annex 3. *Within* and aggregate pseudo-panel models with all variables on the share of expenditure on investment and on total expenditure, 2009-2017

*Within* models and variants with all variables: Effect of access to and use of financial services on the share of expenditure on investment

| Effects on the share of expenditure allocated to investment | | | | | |
|---|---|---|---|---|---|
| | (1) | (2) | (3) | (4) | (5) |
| **Variables** | **Baseline model** | **Loan interaction** | **Disaggr. deposits** | **Squared access** | **HHI** |
| $(0<Dist\leq 20)_t$ | 0,00334*** | 0,00319*** | 0,00343*** | 0,00330*** | 0,00410*** |
| | (0,000658) | (0,000659) | (0,000658) | (0,000659) | (0,000689) |
| $(20<Dist\leq 50)_t$ | 0,00181*** | 0,00166** | 0,00190*** | 0,00177** | 0,00258*** |
| | (0,000689) | (0,000689) | (0,000688) | (0,000689) | (0,000718) |
| Banks $_t$ | -0,000772 | -0,000663 | -0,000739 | -0,000697 | -0,000745 |
| | (0,000535) | (0,000525) | (0,000537) | (0,000820) | (0,000533) |
| Municipal savings banks $_t$ | 0,000913* | 0,000869 | 0,000922* | 0,000811 | 0,000907* |
| | (0,000534) | (0,000531) | (0,000533) | (0,000922) | (0,000537) |
| Banking agents $_t$ | -1,55e-06 | -2,90e-06 | -9,58e-07 | -1,60e-05 | -7,73e-07 |
| | (1,15e-05) | (1,14e-05) | (1,15e-05) | (2,12e-05) | (1,15e-05) |
| Caja loans $_t$ | 0,00291*** | 0,00279*** | 0,00394*** | 0,00296*** | 0,00260*** |
| | (0,000619) | (0,000622) | (0,000770) | (0,000676) | (0,000754) |
| Bank loans $_t$ | -0,00159*** | -0,00247*** | -0,00128*** | -0,00156*** | -0,00122** |
| | (0,000499) | (0,000569) | (0,000494) | (0,000525) | (0,000533) |
| Total deposits t | 0,00262*** | 0,00264*** | | 0,00262*** | 0,00241*** |
| | (0,000415) | (0,000415) | | (0,000415) | (0,000416) |
| User | 0,0105*** | 0,0105*** | 0,0105*** | 0,0105*** | 0,0104*** |
| | (0,00265) | (0,00265) | (0,00265) | (0,00265) | (0,00265) |
| Education level | 0,00413*** | 0,00413*** | 0,00413*** | 0,00413*** | 0,00411*** |
| | (0,000143) | (0,000143) | (0,000143) | (0,000143) | (0,000143) |
| Sex | -0,000324 | -0,000323 | -0,000336 | -0,000329 | -0,000265 |
| | (0,000692) | (0,000692) | (0,000692) | (0,000693) | (0,000693) |

| Effects on the share of expenditure allocated to investment | | | | | |
|---|---|---|---|---|---|
| Married | 0,00461*** | 0,00462*** | 0,00463*** | 0,00462*** | 0,00460*** |
| | (0,000537) | (0,000537) | (0,000537) | (0,000537) | (0,000537) |
| Age | -0,00028*** | -0,00028*** | -0,00028*** | -0,00028*** | -0,00028*** |
| | (7,77e-05) | (7,77e-05) | (7,77e-05) | (7,77e-05) | (7,77e-05) |
| Not working | 0,00927*** | 0,00927*** | 0,00928*** | 0,00927*** | 0,00931*** |
| | (0,000987) | (0,000987) | (0,000987) | (0,000987) | (0,000987) |
| Self-employed | 0,00695*** | 0,00694*** | 0,00694*** | 0,00694*** | 0,00696*** |
| | (0,000584) | (0,000584) | (0,000584) | (0,000584) | (0,000584) |
| Employer | 0,0272*** | 0,0272*** | 0,0272*** | 0,0272*** | 0,0272*** |
| | (0,00118) | (0,00118) | (0,00118) | (0,00118) | (0,00118) |
| (Caja loans $_t$) * | | 0,000575 | | | |
| (Bank loans $_t$) | | (0,000356) | | | |
| Bank deposits $_t$ | | | 0,00284*** | | |
| | | | (0,000445) | | |

Source: Authors' elaboration

*Within* models and variants with all variables: Effect of access to and use of financial services on the share of expenditure on investment (cont.)

| Effects on the share of expenditure allocated to investment | | | | | |
|---|---|---|---|---|---|
| | (1) | (2) | (3) | (4) | (5) |
| **Variables** | **Baseline model** | **Loan interaction** | **Disaggr. deposits** | **Squared access** | **HHI** |
| Caja deposits $_t$ | | | -0,000979 | | |
| | | | (0,00186) | | |
| (Banks $_t$)$^2$ | | | | 1,61e-06 | |
| | | | | (2,12e-05) | |
| (Municipal savings banks $_t$)$^2$ | | | | 5,17e-06 | |
| | | | | (2,88e-05) | |
| (Banking agents $_t$)$^2$ | | | | 8,10e-09 | |
| | | | | (1,21e-08) | |
| High comp. | | | | | -0,000446 |
| | | | | | (0,00152) |
| Medium comp. | | | | | 0,00167 |
| | | | | | (0,00151) |
| Low comp. | | | | | 0,00331*** |
| | | | | | (0,000878) |
| Constant | -8,88e-06 | -0,000135 | -8,89e-06 | -2,89e-05 | -9,13e-06 |
| | (0,000241) | (0,000250) | (0,000241) | (0,000242) | (0,000241) |
| Observations | 103.098 | 103.098 | 103.098 | 103.098 | 103.098 |

Robust standard errors in parentheses

*** p<0,01, ** p<0,05, * p<0,1

Source: Authors' elaboration

*Within* models and variants with all variables: Effect of access to and use of financial services on total expenditure

| Effects on total expenditure | | | | | |
|---|---|---|---|---|---|
| | (1) | (2) | (3) | (4) | (5) |
| **Variables** | **Baseline model** | **Loan interaction** | **Disaggr. deposits** | **Squared access** | **HHI** |
| $(0<Dist\leq 20)_t$ | -0,139*** | -0,135*** | -0,139*** | -0,139*** | -0,132*** |
| | (0,00526) | (0,00624) | (0,00621) | (0,00621) | (0,00667) |
| $(20<Dist\leq 50)_t$ | -0,188*** | -0,184*** | -0,188*** | -0,188*** | -0,181*** |
| | (0,00565) | (0,00676) | (0,00674) | (0,00674) | (0,00718) |
| Banks $_t$ | 0,0238*** | 0,0210*** | 0,0239*** | 0,0812*** | 0,0228*** |
| | (0,00413) | (0,00458) | (0,00457) | (0,00810) | (0,00456) |
| Municipal savings banks $_t$ | -0,0296*** | -0,0285*** | -0,0296*** | -0,151*** | -0,0287*** |
| | (0,00387) | (0,00432) | (0,00432) | (0,00923) | (0,00429) |
| Banking agents $_t$ | 0,00048*** | 0,00051*** | 0,00048*** | 0,00137*** | 0,00046*** |
| | (9,08e-05) | (9,86e-05) | (9,77e-05) | (0,000216) | (9,78e-05) |
| Caja loans $_t$ | 0,0871*** | 0,0900*** | 0,0901*** | 0,132*** | 0,0796*** |
| | (0,00461) | (0,00518) | (0,00668) | (0,00567) | (0,00620) |
| Bank loans $_t$ | -0,0364*** | -0,0135*** | -0,0354*** | -0,0489*** | -0,0399*** |
| | (0,00331) | (0,00490) | (0,00395) | (0,00406) | (0,00407) |
| Total deposits t | 0,0579*** | 0,0573*** | | 0,0581*** | 0,0574*** |
| | (0,00290) | (0,00377) | | (0,00380) | (0,00383) |
| User | 0,289*** | 0,289*** | 0,289*** | 0,289*** | 0,289*** |
| | (0,0182) | (0,0231) | (0,0231) | (0,0231) | (0,0231) |
| Education level | 0,0961*** | 0,0962*** | 0,0961*** | 0,0966*** | 0,0960*** |
| | (0,00111) | (0,00135) | (0,00135) | (0,00135) | (0,00135) |
| Sex | 0,0536*** | 0,0535*** | 0,0535*** | 0,0523*** | 0,0538*** |
| | (0,00581) | (0,00708) | (0,00708) | (0,00707) | (0,00708) |
| Married | 0,189*** | 0,189*** | 0,189*** | 0,190*** | 0,189*** |
| | (0,00429) | (0,00518) | (0,00518) | (0,00517) | (0,00518) |
| Age | -0,0118*** | -0,0118*** | -0,0118*** | -0,0116*** | -0,0118*** |
| | (0,000594) | (0,000753) | (0,000753) | (0,000752) | (0,000753) |
| Not working | -0,0301*** | -0,0302*** | -0,0302*** | -0,0299*** | -0,0298*** |
| | (0,00759) | (0,00920) | (0,00921) | (0,00919) | (0,00921) |
| Self-employed | -0,180*** | -0,180*** | -0,180*** | -0,177*** | -0,180*** |
| | (0,00476) | (0,00574) | (0,00574) | (0,00574) | (0,00574) |
| Employer | 0,211*** | 0,212*** | 0,211*** | 0,210*** | 0,211*** |
| | (0,00814) | (0,00982) | (0,00982) | (0,00980) | (0,00982) |
| (Caja loans $_t$) * (Bank loans $_t$) | | -0,0150*** | | | |
| | | (0,00204) | | | |
| Bank deposits $_t$ | | | 0,0586*** | | |
| | | | (0,00403) | | |
| Caja deposits $_t$ | | | 0,0469*** | | |
| | | | (0,0148) | | |
| (Banks $_t$)$^2$ | | | | -0,00162*** | |
| | | | | (0,000213) | |
| (Municipal savings banks $_t$)$^2$ | | | | 0,00391*** | |
| | | | | (0,000277) | |

Source: Authors' elaboration

*Within* models and variants with all variables: Effect of access to and use of financial services on total expenditure (cont.)

| Effects on total expenditure | | | | | |
|---|---|---|---|---|---|
| | (1) | (2) | (3) | (4) | (5) |
| **Variables** | **Baseline model** | **Loan interaction** | **Disaggr. deposits** | **Squared access** | **HHI** |
| (Banking agents $_t$)$^2$ | | | | -6,4e-07*** | |
| | | | | (1,04e-07) | |
| High comp. | | | | | 0,0534*** |
| | | | | | (0,0137) |
| Medium comp. | | | | | -0,0231* |
| | | | | | (0,0138) |
| Low comp. | | | | | 0,0265*** |
| | | | | | (0,00878) |
| Constant | -0,236*** | 8,852*** | 8,848*** | 8,845*** | 8,848*** |
| | (0,00193) | (0,00239) | (0,00235) | (0,00237) | (0,00235) |
| Observations | 103.098 | 103.098 | 103.098 | 103.098 | 103.098 |

Robust standard errors in parentheses

*** p<0,01, ** p<0,05, * p<0,1

Source: Authors' elaboration

Aggregate models with all variables: Effect of access to and use of financial services on the share of expenditure on investment and on total expenditure

| | Effect on the share of expenditure allocated to investment | | Effect on total expenditure | |
|---|---|---|---|---|
| | (6,1) | (7,1) | (6,2) | (7,2) |
| **Variables** | **Ubigeo model** | **Ubigeo and language model** | **Ubigeo model** | **Ubigeo and language model** |
| $(0 < Dist \leq 20)_t$ | 0,00307*** | 0,00342*** | -0,0748*** | -0,0967*** |
| | (0,00116) | (0,00121) | (0,0141) | (0,0142) |
| $(20 < Dist \leq 50)_t$ | 0,00234* | 0,00274** | -0,143*** | -0,146*** |
| | (0,00124) | (0,00131) | (0,0159) | (0,0158) |
| Caja loans $_t$ | 0,00115 | -7,65e-07 | 0,0926*** | 0,102*** |
| | (0,00102) | (0,00112) | (0,0140) | (0,0130) |
| Bank loans $_t$ | -0,00104 | -3,39e-05 | -0,0155* | -0,0384*** |
| | (0,000842) | (0,00102) | (0,00927) | (0,0102) |
| Banks $_t$ | -0,000330 | -0,000849 | 0,00671 | 0,0207* |
| | (0,00136) | (0,00142) | (0,0117) | (0,0118) |
| Municipal savings banks $_t$ | 0,000958 | 0,00140 | -0,0159 | -0,0188** |
| | (0,00128) | (0,00136) | (0,0109) | (0,00922) |
| Banking agents $_t$ | -7,97e-06 | -1,25e-05 | 0,000473** | 0,000206 |
| | (1,33e-05) | (1,51e-05) | (0,000215) | (0,000162) |
| Total deposits $_t$ | 0,00201*** | 0,00217*** | 0,0328*** | 0,0512*** |
| | (0,000545) | (0,000620) | (0,00868) | (0,0105) |
| Quechua | 0,00555*** | 0,00504*** | -0,148*** | -0,0802*** |
| | (0,00125) | (0,00117) | (0,0165) | (0,0144) |
| Other languages | 0,000574 | -0,00287 | -0,396*** | -0,269*** |
| | (0,00210) | (0,00189) | (0,0311) | (0,0276) |
| User | 0,00188** | 0,00171* | 0,137*** | 0,132*** |
| | (0,000922) | (0,000980) | (0,0116) | (0,0117) |
| Education level | 0,00397*** | 0,00468*** | 0,214*** | 0,172*** |

| | Effect on the share of expenditure allocated to investment | | Effect on total expenditure | |
|---|---|---|---|---|
| | (0,000623) | (0,000586) | (0,00753) | (0,00666) |
| Sex | -0,00383 | -0,000787 | -0,0814* | -0,0291 |
| | (0,00402) | (0,00409) | (0,0495) | (0,0459) |
| Married | 0,00335 | 0,00186 | 0,111*** | 0,0802*** |
| | (0,00256) | (0,00263) | (0,0322) | (0,0306) |
| Age | -0,000353*** | -0,000435*** | -0,00423*** | -0,00138 |
| | (8,95e-05) | (8,91e-05) | (0,00106) | (0,000998) |
| Not working | 0,00854 | 0,0109* | -0,144** | -0,183*** |
| | (0,00538) | (0,00592) | (0,0576) | (0,0571) |
| Self-employed | 0,00130 | 0,00360 | -0,741*** | -0,763*** |
| | (0,00292) | (0,00275) | (0,0342) | (0,0327) |
| Employer | 0,0389*** | 0,0398*** | -0,100 | -0,0555 |
| | (0,00549) | (0,00597) | (0,0669) | (0,0628) |
| Constant | 0,0713*** | 0,0683*** | 8,703*** | 8,748*** |
| | (0,00596) | (0,00615) | (0,0696) | (0,0682) |
| Observations | 4.487 | 5.037 | 4.487 | 5.037 |

Robust standard errors in parentheses

*** p<0,01, ** p<0,05, * p<0,1

Source: Authors' elaboration